http://pubsonline.informs.org/journal/mnsc

MANAGEMENT SCIENCE
Vol. 67, No. 1, January 2021, pp. 220–241
ISSN 0025-1909 (print), ISSN 1526-5501 (online)

Small-Data, Large-Scale Linear Optimization with Uncertain Objectives

Vishal Gupta,[a] Paat Rusmevichientong[a]

[a] Data Science and Operations, University of Southern California Marshall School of Business, Los Angeles, California 90089

Contact: guptavis@usc.edu, https://orcid.org/0000-0003-4371-9114 (VG); rusmevic@marshall.usc.edu, https://orcid.org/0000-0001-9584-4203 (PR)

Received: November 6, 2017
Revised: February 27, 2019; October 1, 2019
Accepted: November 18, 2019
Published Online in Articles in Advance: July 1, 2020

https://doi.org/10.1287/mnsc.2019.3554

Copyright: © 2020 INFORMS

Abstract. Optimization applications often depend on a huge number of uncertain parameters. In many contexts, however, the amount of relevant data per parameter is small, and hence, we may only have imprecise estimates. We term this setting—in which the number of uncertainties is large but all estimates have low precision—the *small-data, large-scale regime*. We formalize a model for this new regime, focusing on optimization problems with uncertain linear objectives. We show that common data-driven methods, such as sample average approximation, data-driven robust optimization, and certain regularized policies, may perform poorly in this new setting. We then propose a novel framework for selecting a data-driven policy from a given policy class. As with the aforementioned data-driven methods, our new policy enjoys provably good performance in the large-sample regime. Unlike these methods, we show that in the small-data, large-scale regime, our data-driven policy performs comparably to an oracle best-in-class policy under some mild conditions. We strengthen this result for linear optimization problems and two natural policy classes, the first inspired by the empirical Bayes literature and the second by regularization techniques. For both classes, the suboptimality gap between our proposed policy and the oracle policy decays exponentially fast in the number of uncertain parameters even for a fixed amount of data. Thus, these policies retain the strong large-sample performance of traditional methods and additionally enjoy provably strong performance in the small-data, large-scale regime. Numerical experiments confirm the significant benefits of our methods.

History: Accepted by Yinyu Ye, optimization.
Funding: This work was supported by the National Science Foundation, Division of of Civil, Mechanical and Manufacturing Innovation [Grants CMMI-1661732 and CMMI-1824860].
Supplemental Material: The online appendices are available at https://doi.org/10.1287/mnsc.2019.3554.

Keywords: data-driven optimization • small-data • large-scale regime • Stein's unbiased risk estimation

1. Introduction

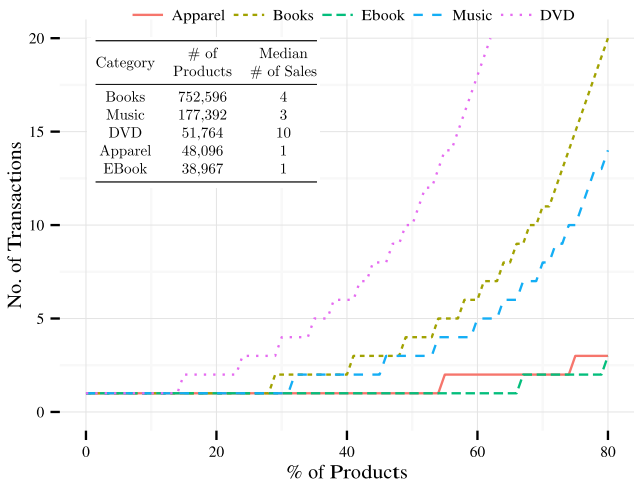
We live in a small-data world. Popular press about the age of "big data" notwithstanding, many real-world decision-making problems exhibit a huge number of uncertain parameters with a small amount of relevant data per parameter. Consider the following two examples.

Inventory Management for Low-Demand Products. Large online retailers carry millions of products, but most products have few sales per quarter. Figure 1 summarizes the number of sales per product for the five most popular product categories based on data for a large e-retailer. The vast majority of products have fewer than five sales. Consequently, although there are *many* products with uncertain demands, there are *limited* demand data per product.

Vehicle Routing After an Accident. Ride-sharing platforms, such as Uber, and logistics services, such as UPS, frequently update routing decisions based on near real-time traffic data across millions of road segments. In the wake of a large accident or weather disruption, however, historical traffic patterns often shift dramatically; there may only be a few hours of relevant data on the new, postdisruption travel times. Again, although there are *many* road segments with uncertain travel times, there are *limited* traffic data per segment.

Some reflection suggests that the small-amount-of-relevant-data phenomenon in these examples is at least partially driven by the nature of modern decision making under uncertainty. It is not unusual for real applications to require making thousands of decisions simultaneously in time-changing environments with only low-precision estimates. This combination of features—highly granular decision making, time-changing environments, and low-precision estimates—drives the small-data, large-scale phenomenon. Their ubiquity suggests that a host of other applications, such as new-user product recommendations and disaster response operations, may also exhibit these features. We term this decision-making setting—many

Figure 1. (Color online) Number of Sales per Product



Category	# of Products	Median # of Sales
Books	752,596	4
Music	177,392	3
DVD	51,764	10
Apparel	48,096	1
EBook	38,967	1

Notes. Data taken from a leading internet retailer for all products that sold at least one unit in the top five categories between July and September 2005. Except for the DVD category, the median number of sales for each product is at most five.

uncertain parameters, each with limited relevant data and, hence, an imprecise estimate—the *small-data, large-scale regime*.

By contrast, many traditional data-driven optimization methods are theoretically justified by studying their performance in the large-sample regime, in which the number of uncertain parameters is fixed, but we have access to increasing amounts of data and, hence, increasingly precise estimates of all parameters. For example, the sample average approximation (SAA) approach is well known to converge to the full-information optimal performance in the large-sample regime (Kleywegt et al. 2002, Shapiro et al. 2009). This type of large-sample performance guarantee shows that these methods perform well when the amount of data is large relative to the number of uncertainties. However, it is unclear how these methods may perform in the small-data, large-scale regime.

At a high level, the main messages of this paper are as follows. First, the small-data, large-scale regime is structurally different from the large-sample regime, and consequently, traditional methods may perform quite poorly in this regime. Second, it is possible to design novel methods that retain the strong large-sample performance guarantees of traditional methods but that *additionally* have provably good and empirically strong performance in the small-data, large-scale regime.

Because the class of all small-data, large-scale optimization problems is too broad to treat in a single paper, we focus on problems with known feasible regions but an uncertain, linear objective.

$$\mathrm{P}^n: \quad Z^*(\boldsymbol{\mu}) \equiv \max_{x\in\mathcal{X}} \frac{1}{n}\boldsymbol{\mu}^\top x.$$

Here $\mathcal{X} \subseteq \mathbb{R}^n$ is a known, nonempty, compact, potentially nonconvex set, and the objective coefficients $\boldsymbol{\mu}$ are unknown. Instead, we are given noisy estimates $\hat{\mu}_j$ for each j, each of which was formed using only a small amount of relevant data. (A precise model for $\hat{\boldsymbol{\mu}}$ is given in Section 2.) Let $x^*(\boldsymbol{\mu})$ denote an optimal solution to P^n.

Admittedly, problem P^n is not general enough to cover sophisticated models for the aforementioned inventory-management and vehicle-routing applications. That said, we do consider P^n to be a fundamental building block for these applications. In particular, P^n does subsume transportation and shortest-path problems as special cases (Bertsimas and Tsitsiklis 1997, chapter 7).

Our Contributions and Main Results. Using problem P^n, we highlight unique features and challenges of the small-data, large-scale regime. In particular, in contrast to the large-sample regime, the classical SAA method can perform arbitrarily badly (Example 3), and no data-driven optimization procedure can guarantee more than a constant fraction of the full-information optimal value $Z^*(\boldsymbol{\mu})$ (Theorem 2.7).

Because consistently attaining full-information performance is provably impossible in the small-data, large-scale regime by Theorem 2.7, we instead restrict attention to classes of data-driven policies and seek methods for identifying a member whose performance is comparable with a certain oracle policy. This oracle has access to the true value of $\boldsymbol{\mu}$ in P^n but is restricted to use a policy from the class, thus formalizing the notion of a best-in-class policy.

We propose a novel framework for this task and show that if the given policy class satisfies a certain uniform convergence criterion given in Theorem 3.5, then our proposed policy performs comparably to the oracle policy. Specifically, we prove that when the estimates $\hat{\mu}_j$ are Gaussian and $n \to \infty$ in P^n, the performance of our proposed policy converges to that of the oracle policy even if the amount of relevant data per uncertainty remains fixed. Moreover, when $\hat{\mu}_j$ have non-Gaussian distributions, the difference in performance between our proposed policy and the oracle policy converges to a constant that measures the degree of nonnormality, and this constant does not depend on the parameters of P^n.

We then specialize and strengthen our general result to linear optimization problems, that is, when $\mathcal{X}$ is polyhedral (cf. Equation (4.1)), focusing on two specific policy classes. The first, which we call *Bayes-inspired policies*, is motivated by the empirical Bayes and compound estimation literature in statistics (see Zhang 2003, Efron 2012, and the references therein). These methods target applications such as microarray

analysis, genomics, and compressed sensing, all of which involve simultaneously solving thousands of separate inference problems with limited data. Thus, these methods are particularly suited for the small-data, large-scale regime. Indeed, simple estimate-then-optimize policies leveraging empirical Bayes' estimators already outperform SAA in the small-data, large-scale regime (cf. Section 4). However, we observe empirically that these policies do not typically achieve near-oracle performance because they fail to exploit the particular optimization structure. By contrast, by specializing our general framework, we propose a new policy that achieves near-oracle performance for large n and, thus, necessarily outperforms these estimate-then-optimize variants. We strengthen our previous general-purpose results by providing an explicit, nonasymptotic bound on the suboptimality gap that converges to the aforementioned *nonnormality* constant. This bound converges exponentially fast in the number of uncertain parameters n even for a fixed amount of data per parameter (Theorem 4.3).

The second policy class that we consider, which we call *regularization-inspired policies*, is motivated by the growing literature on incorporating regularizers into the SAA problem to improve performance and computational tractability (see Nesterov 2005, Negahban et al. 2012, and the references therein). We focus on a weighted ℓ_2-regularizer. Based on a well-known equivalence between regularization and robustness, these policies are equivalent to policies obtained by solving a robust optimization problem with an ellipsoidal uncertainty set (Lemma E.1 in Online Appendix E). We prove that common cross-validation techniques—which are routinely used to specify the regularization parameter in the large-sample regime—do not achieve near-oracle performance (Theorem 5.1). Similarly, we illustrate empirically that policies based on probabilistic feasibility guarantees—a standard approach to sizing uncertainty sets in robust optimization—also do not achieve near-oracle performance. By contrast, by specializing our general framework, we propose a new policy that achieves near-oracle performance for large n and, thus, may outperform these approaches. We again improve on our previous general-purpose result by proving a nonasymptotic bound on its suboptimality gap that converges exponentially fast to zero (if the $\hat{\mu}_j$ are Gaussian) and converges exponentially fast to the aforementioned nonnormality constant (Theorem 5.2) otherwise.

In all three of our small-data, large-scale performance guarantees, the suboptimality of our policy depends on the degree of nonnormality of the estimators $\hat{\mu}_j$. Methods that either explicitly or implicitly assume Gaussian inputs are common in the high-dimensional statistics and compound decision-making literature, at least partially because many simple estimators $\hat{\mu}_j$ are approximately normally distributed (see Examples 2.2 and 2.5). Consequently, many authors derive estimation procedures that are provably optimal under Gaussian assumptions and then argue that such procedures have good *practical* performance even when strict normality does not hold (see, e.g., Morris 1983; Efron and Morris 1973, 1975; Donoho and Johnstone 1995; Xie et al. 2012; Mukherjee et al. 2015). We adopt a similar perspective. Our bounds make theoretically precise the sense in which our policy can be expected to perform well as normality is violated, and in Section F.5 of Online Appendix F, we provide a preliminary empirical assessment of our methods under nonnormality.

Importantly, we stress that our small-data, large-scale performance guarantees for our proposed policies come at essentially *no statistical cost* in the large-sample regime. Specifically, we prove an explicit, nonasymptotic suboptimality bound relative to the full-information optimum $Z_n^*(\boldsymbol{\mu})$ for our policy (Theorem 3.6). In the large-sample setting with independent and identically distributed (i.i.d.) observations, this bound converges to zero at a rate comparable to SAA. In practice, the performance may even be much better than SAA (see, e.g., Section F.4 of Online Appendix F). In this sense, we argue that our methods retain the strong large-sample properties of SAA and also enjoy additional small-data, large-scale performance guarantees.

Finally, we present a simulation study calibrated to real data for managing a portfolio of online advertisements. Our methods outperform traditional methods and estimate-then-optimize methods from high-dimensional statistics so long as the number of uncertainties is sufficiently large.

Relationship to Prior Work. Problem P^{n} is a special case of the stochastic optimization problem

$$\min_{\boldsymbol{x}\in\mathscr{X}\subseteq\mathbb{R}^n} \quad \mathbb{E}[c(\boldsymbol{x},\boldsymbol{\xi})], \tag{1.1}$$

in which $c(\boldsymbol{x},\boldsymbol{\xi}) = \frac{1}{n}\boldsymbol{\xi}^\top\boldsymbol{x}$ and $\mathbb{E}[\boldsymbol{\xi}] = \boldsymbol{\mu}$. There is now a rich literature in data-driven optimization on solving data-driven versions of Equation (1.1) when the distribution of $\boldsymbol{\xi}$ is unknown, but one has access to a data set $\hat{\boldsymbol{\xi}}^1,\ldots,\hat{\boldsymbol{\xi}}^{\mathrm{S}}$ drawn i.i.d. as $\boldsymbol{\xi}$.[1] Our model for $\hat{\boldsymbol{\mu}}$ approximates this data-generation mechanism (cf. Example 2.2) and also captures other settings (e.g., Example 2.5).

As mentioned, many methods for addressing data-driven versions of Equation (1.1) satisfy large-sample performance guarantees similar to those of SAA. Such methods include robust SAA (Bertsimas et al. 2018b), regularized SAA variants (Negahban et al. 2012), stochastic gradient descent and its variants (Nemirovski et al. 2009, Nesterov 2009, Lan 2012), and distributional robust optimization (Delage and

Ye 2010, Esfahani and Kuhn 2018, Gupta 2019). Many of these methods seek to improve on SAA by establishing guarantees beyond large-sample performance. For example, many data-driven robust and distributionally robust optimization methods can be tuned to ensure that for finite data, solutions satisfy certain probabilistic feasibility guarantees (see Bertsimas et al. 2018a, b) for recent references). Meanwhile, gradient-based methods with suitable step sizes can offer probabilistic optimality guarantees for finite samples. Finally, certain regularizers are known to yield either consistent estimates of the unknown parameter or consistent predictions if the underlying parameter satisfies a certain a priori structure, such as sparsity (see Bickel et al. (2009), Candès and Recht (2009), and Wainwright (2009) for representative results and Negahban et al. (2012) for additional references). These results do not directly extend to the small-data, large-scale regime.

There is also a separate literature studying Equation (1.1) in the high-dimensional statistics and machine learning communities. In this setting, ξ is often, but not always, an $n+1$-dimensional vector consisting of an n-dimensional feature component and a one-dimensional response component, and Equation (1.1) represents a prediction problem. Most relevantly for our work, this literature has placed special emphasis on studying the performance of methods under various asymptotic scalings of n (the number of decision variables) and S (the number of samples), especially the large-sample regime ($\mathsf{S} \to \infty$ and n fixed) and the high-dimensional regime (both $\mathsf{S} \to \infty$ and $n \to \infty$). The celebrated Vapnik–Chervonenkis (VC)–dimension theory (Vapnik 1999) and algorithm stability theory (Bousquet and Elisseeff 2002) both provide general frameworks for analyzing approaches to Equation (1.1) in these regimes. Many works exist that strengthen these performance guarantees for special cases of Equation (1.1) (see, e.g., Bühlmann and Geer (2011) and references therein for lasso regression and Belloni and Chernozhukov (2011) for penalized quantile regression). Ban and Rudin (2014) connect this literature with the operations literature, highlighting the benefits of leveraging feature vectors in the particular case of the newsvendor problem in both the large-sample and high-dimensional regimes. Subsequently, other authors have explored incorporating feature information in operations research problems in terms of both developing high-quality estimates to plug into an optimization formulation (Ferreira et al. 2015, Chen et al. 2015, Ban et al. 2018) and approaches that blend estimation and optimization (Elmachtoub and Grigas 2017, Bertsimas and Kallus 2019).

One can view our work as complementing this statistical literature in the setting in which S is fixed and $n \to \infty$ (again, see, Example 1), although all our results are stated in a nonasymptotic framework. From a technical point of view, although we leverage some standard tools, for example, pseudodimension (an extension of VC dimension), the structure of P^{n} requires novel and specialized analysis, especially for the polyhedral feasibility sets considered in Sections 4 and 5. One cannot simply apply the VC theory "out of the box" because these constraints introduce a nontrivial dependence structure, and the requisite quantities are thus *not* simple averages of independent, random functions. Moreover, we focus on proving best-in-class optimality results, although much of the high-dimensional literature instead focuses on generalization results that bound the difference between in-sample and out-of-sample performance.

Our approach to computing asymptotically best-in-class policies uses a novel adaptation of Stein's unbiased risk estimation (SURE; Stein 1981), a common approach to model selection in statistics (for representative examples, see Donoho and Johnstone (1995), Tibshirani and Taylor (2012), Candes et al. (2013)). To the best of our knowledge, however, SURE has never been leveraged for more general data-driven optimization problems. The key idea of our adaptation is to replace the standard SURE estimator with an asymptotically unbiased approximation constructed so that the approximation error vanishes as the number of uncertainties grows large. In this sense, our approach is similar in spirit to that of Mukherjee et al. (2015), who also develop an asymptotic approximation to SURE. However, Mukherjee et al. (2015) heavily leverage the specific structure of the check-loss function, and our approach applies generally to optimization problems with linear objectives as long as the class of policies satisfies the requisite uniform convergence criteria.

2. Formulation and Properties of the Small-Data, Large-Scale Regime

Throughout the remainder of this paper, vectors are denoted in bold, and scalars appear in a regular font. We use $\phi(\cdot)$ and $\Phi(\cdot)$ to denote the density and cumulative distribution functions of the standard normal random variable. We write $\hat{\mu}_j \sim \mathcal{N}(m, v)$ to indicate that the random variable $\hat{\mu}_j$ is normally distributed with mean m and variance v.

Importanty, we adopt the following assumption on the estimates $\hat{\mu}_j$.

Assumption 2.1 (Model for $\hat{\boldsymbol{\mu}}$). *For each $j = 1, \ldots, n$, $\hat{\mu}_j$ is unbiased, that is, $\mathbb{E}[\hat{\mu}_j] = \mu_j$, and has known precision ν_j, that is, $\mathbb{E}[(\hat{\mu}_j - \mu_j)^2] = 1/\nu_j$.*

Recall that precision is the reciprocal of the variance. Similar assumptions, that is, mean-zero noise with some measure of dispersion, are very common in the high-dimensional statistics literature (Johnstone 2015)

and are often used in the robust optimization literature to assess the performance of uncertainty sets (see, e.g., Ben-Tal and Nemirovski 2000, Bertsimas and Sim 2004, Chen et al. 2007). The precisions ν_j implicitly measure the amount of relevant data used in constructing $\hat{\mu}_j$. We use them to define the small-data, large-scale regime in general settings. As a first step, consider the following motivating example.

Example 2.2 (Finite Observation Model)**.** Suppose that for each $j = 1, \ldots, n$, we observe $\mathsf{S}_j \geq 1$ i.i.d. random variables $\xi_j^1, \ldots, \xi_j^{\mathsf{S}_j}$, each with mean μ_j and known precision ν_0. Because the μ_j are potentially unrelated and all draws are independent, the data $\xi_j^1, \ldots, \xi_j^{\mathsf{S}_j}$ provide no information for estimating μ_k whenever $k \neq j$. Thus, the sample mean $\hat{\mu}_j = \frac{1}{\mathsf{S}_j}\sum_{\ell=1}^{\mathsf{S}_j} \xi_j^\ell$ is arguably the most natural, unbiased estimate of μ_j and has precision $\nu_j \equiv \mathsf{S}_j\nu_0$.

In this example, the small-data, large-scale regime corresponds to the setting in which n is very large *but* S_j is fixed and small for each j. In particular, although the total amount of data $\sum_{j=1}^n \mathsf{S}_j \to \infty$ as $n \to \infty$, the precision ν_j remains bounded for each j. Thus, the additional data do not provide additional precision in estimating μ_j, that is, $\hat{\mu}_j \not\to_p \mu_j$ as $n \to \infty$. In this sense, there is a small amount of relevant data for estimating each μ_j. By contrast, the large-sample regime in this example corresponds to the setting in which n is fixed and S_j is large for each j. In particular, for each j, $\nu_j \to \infty$ as $\mathsf{S}_j \to \infty$ and $\hat{\mu}_j \to_p \mu_j$ as $\mathsf{S}_j \to \infty$ by the law of large numbers.[2]

Note the distinct behavior of the ν_j in these two regimes. Our definition of the small-data, large-scale regime is inspired by this difference. Specifically, for any $\hat{\boldsymbol{\mu}} \in \mathbb{R}^n$ and $\boldsymbol{\nu} \in \mathbb{R}_+^n$, let the tuple $(\mathrm{P}^n, \hat{\boldsymbol{\mu}}, \boldsymbol{\nu})$ denote an instance of P^n with unbiased estimators $\hat{\boldsymbol{\mu}} = (\hat{\mu}_1, \ldots, \hat{\mu}_n)^\top$ with corresponding precisions $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_n)^\top$.

Definition 2.3 (Small-Data, Large-Scale Regime)**.** We say that a sequence of instances $\{(\mathrm{P}^n, \hat{\boldsymbol{\mu}}^n, \boldsymbol{\nu}^n) : \hat{\boldsymbol{\mu}}^n \in \mathbb{R}^n, \boldsymbol{\nu}^n \in \mathbb{R}_+^n, n \geq 2\}$ is in the *small-data, large-scale regime* if there exists a $\nu_{\max} < \infty$ such that $\nu_j^n \leq \nu_{\max}$ for all $j = 1, \ldots, n$ and $n = 2, \ldots, \infty$.[3]

In other words, in the small-data, large-scale regime, all estimates have bounded precision. For comparison, we also define the large-sample regime. In this case, we consider a sequence of estimators that are improving in quality as we collect more data. This sequence requires an extra indexing variable S (mnemonically, S stands for the "number of samples").

Definition 2.4 (Large-Sample Regime)**.** For a fixed n, we say a sequence of instances $\{(\mathrm{P}^n, \hat{\boldsymbol{\mu}}^{\mathsf{S}}, \boldsymbol{\nu}^{\mathsf{S}}) : \hat{\boldsymbol{\mu}}^{\mathsf{S}} \in \mathbb{R}^n, \boldsymbol{\nu}^{\mathsf{S}} \in \mathbb{R}_+^n, \mathsf{S} \geq 1\}$ is in the *large-sample regime* if $\lim_{\mathsf{S}\to\infty} \min_{j=1,\ldots,n} \nu_j^{\mathsf{S}} = \infty$.

We stress that in the large-sample regime, the problem P^n is fixed, but the precision of each estimate $\hat{\boldsymbol{\mu}}^{\mathsf{S}}$ increases with S. In contrast, in the small-data, large-scale regime, the dimension of the problem P^n increases with n while the precision remains bounded. We prefer modeling in terms of precisions instead of number of samples S because it allows us to extend the idea of a "small amount of relevant data" to other settings in a unified way. Consider the following example.

Example 2.5 (Linear Regression)**.** Suppose that for each j, we observe $(\xi_j, \boldsymbol{f}_j) \in \mathbb{R}^{p+1}$, where $\mathbb{E}[\xi_j] = \mu_j$, and $\boldsymbol{f}_j \in \mathbb{R}^p$ is an auxiliary feature vector. For example, in our vehicle-routing example, $\boldsymbol{f}_j$ may capture the speed limit, number of lanes, and distance of the road segment. In such a setting, we might posit a linear regression model, that is,

$$\xi_j = \boldsymbol{\beta}^T \boldsymbol{f}_j + \epsilon_j, \quad \text{with } \mathbb{E}[\epsilon_j] = 0 \text{ and}$$
$$\mathbb{E}\left[\epsilon_j^2\right] = \sigma^2, \quad j = 1, \ldots, n,$$

and then estimate $\boldsymbol{\beta}$ by $\hat{\boldsymbol{\beta}}$ obtained by ordinary least squares. We can map this example to our model by letting $\hat{\mu}_j = \hat{\boldsymbol{\beta}}^\top \boldsymbol{f}_j$. Standard regression results show that $\hat{\mu}_j$ is unbiased with precision $1/\nu_j = \sigma^2 \boldsymbol{f}_j^\top (\boldsymbol{F}^\top \boldsymbol{F})_{jj}^{-1} \boldsymbol{f}_j$, where $\boldsymbol{F} \in \mathbb{R}^{n\times p}$ is the matrix with rows given by $\boldsymbol{f}_j^\top$.

Intuitively, some components of μ_j may have a smaller amount of relevant data because, if some features are very rare, coordinates j depending on these features have less relevant data. The small-data, large-scale regime intuitively corresponds to settings in which n is large and most coordinates depend on a small number of rare features. Making this idea formal in terms of n and p requires additionally specifying structure on the matrix $\boldsymbol{F}$, for example, its eigenspectrum. By contrast, the precisions $\boldsymbol{\nu}_j$ provide a simple way to define the small-data, large-scale regime and large-sample regime for such examples.[4]

Finally, we define a *data-driven policy* $\boldsymbol{x}(\cdot)$ for P^n to be a function mapping $(\hat{\boldsymbol{\mu}}, \boldsymbol{\nu}, \mathscr{X})$ to a feasible solution $\boldsymbol{x}(\hat{\boldsymbol{\mu}}, \boldsymbol{\nu}, \mathscr{X}) \in \mathscr{X}$. To simplify notation, we typically only emphasize the dependence on the random variable $\hat{\boldsymbol{\mu}}$, writing $\boldsymbol{x}(\hat{\boldsymbol{\mu}})$ instead of $\boldsymbol{x}(\hat{\boldsymbol{\mu}}, \boldsymbol{\nu}, \mathscr{X})$. Implicitly, $\boldsymbol{x}(\hat{\boldsymbol{\mu}})$ depends on n.

We are interested in studying the performance of various data-driven policies under both regimes. In the remainder of this section, we highlight the significant differences between these regimes by proving that certain fundamental results for data-driven optimization in the large-sample regime do *not* carry over to the small-data, large-scale regime for P^n.

2.1. The Performance of SAA in the Small-Data, Large-Scale Regime

The SAA method is perhaps the most ubiquitous and well-studied approach to data-driven optimization.

In our context, the SAA policy proxies the unknown $\boldsymbol{\mu}$ by its unbiased estimate $\hat{\boldsymbol{\mu}}$ and optimizes against this proxy.

$$\boldsymbol{x}^{SAA}(\hat{\boldsymbol{\mu}}) \in \arg\max_{\boldsymbol{x}\in\mathscr{X}} \frac{1}{n}\hat{\boldsymbol{\mu}}^\top \boldsymbol{x}$$

(compare also with Example 1). Under mild assumptions in the large-sample regime, $\boldsymbol{x}^{SAA}(\hat{\boldsymbol{\mu}}) \to \boldsymbol{x}^*(\boldsymbol{\mu})$ almost surely (Kleywegt et al. 2002). Other authors, for example, Lim et al. (2011), have shown the fragility of the SAA method in other contexts. We demonstrate that in the small-data, large-scale regime, SAA can perform *arbitrarily* poorly.

Example 2.6 (SAA Performs Poorly in the Small-Data, Large-Scale Regime)**.** Fix any $0 < \alpha < 0.075$, and let $\mathscr{X} = \{\boldsymbol{x} \in [0,1]^n : \frac{1}{n}\sum_{j=1}^n x_j \le \alpha\}$. Let $\hat{\mu}_j \sim \mathcal{N}(\mu_j, 1/\nu_j)$, independently, with

$$\mu_j = \begin{cases} 0 & \text{if } j \text{ is odd,} \\ 1 & \text{if } j \text{ is even,} \end{cases} \qquad \nu_j = \begin{cases} 1 & \text{if } j \text{ is odd,} \\ \nu & \text{if } j \text{ is even.} \end{cases}$$

In words, we would like to identify αn high-reward items using only the noisy estimates $\hat{\mu}_j$. The full-information optimal value is α; that is, $Z_n^*(\boldsymbol{\mu}) = \alpha$.

By inspection, $x_j^{SAA} = \mathbb{I}(\hat{\mu}_j > q^n)\mathbb{I}(\hat{\mu}_j \ge 0)$, where q^n is the $\lfloor n\alpha \rfloor$th largest value among $\hat{\mu}_1, \ldots, \hat{\mu}_n$ except possibly for one fractional component. Note q^n is the $(1-\alpha)$th quantile of the empirical distribution function $q^n = \inf\{x : \frac{1}{n}\sum_{j=1}^n \mathbb{I}(\hat{\mu}_j \le x) \ge 1-\alpha\}$. Because the empirical distribution converges uniformly to the true distribution, the sample quantile q^n also converges to the true quantile (Van der Vaart 2000, lemma 21.2), that is, $q^n \to q_\nu$ as $n \to \infty$, where q_ν solves

$$\begin{aligned} &\frac{1}{2}\Pr\left\{1 + \frac{\zeta}{\sqrt{\nu}} \le q_\nu\right\} + \frac{1}{2}\Pr\{\zeta \le q_\nu\} = 1-\alpha \\ \iff\ &\frac{1}{2}\Phi\big((1-q_\nu)\sqrt{\nu}\,\big) + \frac{1}{2}\Phi(-q_\nu) = \alpha \\ \iff\ &\Phi\big((1-q_\nu)\sqrt{\nu}\,\big) + \Phi(-q_\nu) = 2\alpha, \end{aligned}$$

and ζ is a standard normal random variable. Consider the function $f_\nu(q) = \Phi((1-q)\sqrt{\nu}\,) + \Phi(-q)$. The function $f_\nu(\cdot)$ is strictly decreasing with $f_\nu(-\infty) = 2$ and $f_\nu(\infty) = 0$. Note that $f_\nu(1.01) = \Phi(-0.01\sqrt{\nu}) + \Phi(-1.01) \ge \Phi(-1.01) > 0.1562 > 2\alpha$, where the last inequality follows because $\alpha < 0.075$. Because ν is arbitrary, this means that $q_\nu > 1.01$ for all $\nu > 0$.

Thus, the SAA performance satisfies

$$\begin{aligned} \frac{1}{n}\sum_{j=1}^n \mu_j x_j^{SAA} &= \frac{1}{2}\frac{2}{n}\sum_{k=1}^{n/2} \mathbb{I}(\hat{\mu}_{2k} \ge q^n)\mathbb{I}(\hat{\mu}_j \ge 0) \\ &\to_{(\text{a.s.})} \frac{1}{2}\Phi\big((1-q_\nu)\sqrt{\nu}\,\big), \qquad \text{as } n \to \infty. \end{aligned}$$

Note that $0 \le \frac{1}{2}\Phi((1-q_\nu)\sqrt{\nu}) \le \frac{1}{2}\Phi(-0.01\sqrt{\nu})$ because $q_\nu > 1.01$. Therefore, as $\nu \to \infty$, the preceding limit converges to zero. Thus, in the small-data, large-scale limit with large enough ν, the SAA solution has performance close to zero. For comparison, randomly choosing αn of the indices has expected performance $\alpha/2 > 0$. Moreover, even for finite n and reasonably small ν, the performance can be quite bad. For example, when $\nu = 2$, $n = 100$, and $\alpha = 0.05$, SAA achieves about 80% of the full-information optimum, and our proposed method from Section 4 achieves about 98%.

It may not be surprising that SAA performs badly in the preceding example because it does not leverage any information about the ν_j. Because the odd items have a lower precision (higher variance) than the even items, the $\hat{\mu}_j$ with odd j frequently appear better than the $\hat{\mu}_j$ with even j despite having a lower mean. Other authors have observed similar poor performance of SAA in a finite sample, inspiring techniques that use the ν_j, such as robust optimization and regularization, but these approaches typically do not directly leverage the scale of the optimization. In Section 5, we discuss refinements of these approaches for the small-data, large-scale regime.

2.2. Full-Information Performance Is Unachievable in the Small-Data, Large-Scale Regime

The main result of this section is as follows.

Theorem 2.7 (Unattainability of Full-Information Optimality)**.** *Let $\boldsymbol{x}(\hat{\boldsymbol{\mu}})$ denote any data-driven policy. Then, for each $n \ge 2$, there exist instances of P^n with $\mathscr{X} = [0,1]^n$, $\boldsymbol{\mu} \in \{-1,+1\}^n$, $\hat{\mu}_j \sim \mathcal{N}(\mu_j, 1)$ for all j, and $\hat{\mu}_1, \ldots, \hat{\mu}_n$ are independent such that*

$$\frac{\mathbb{E}\big[\frac{1}{n}\boldsymbol{\mu}^\top \boldsymbol{x}(\hat{\boldsymbol{\mu}})\big]}{Z^*(\boldsymbol{\mu})} < 0.842.$$

This upper bound is not tight. Nonetheless, a consequence of the theorem is that given any data-driven policy $\boldsymbol{x}(\cdot)$, one can construct a sequence of instances $\{(\mathrm{P}^n, \hat{\boldsymbol{\mu}}^n, \boldsymbol{\nu}^n) : n \ge 2\}$ in the small-data, large-scale regime such that the expected performance of $\boldsymbol{x}(\cdot)$ is at most a constant fraction of the full-information optimal performance for each instance.

See Online Appendix B for a proof. Loosely, the proof proceeds by generating random instances of P^n by sampling $\boldsymbol{\mu}$ uniformly from $\{-1,+1\}^n$. The worst-case expected performance of $\boldsymbol{x}(\hat{\boldsymbol{\mu}})$ across these instances is bounded by its average expected performance across these instances. To complete the theorem, we compute an upper bound on this average expected performance. We note that the constant 0.842 arises from the way in which we generate the random instance in the proof. In particular, a different sampling procedure for $\boldsymbol{\mu} \in \{-1,+1\}^n$ might yield tighter upper bounds.

3. Selecting Policies in the Small-Data, Large-Scale Regime

In light of Theorem 2.7, constructing policies that consistently achieve full-information optimal performance in the small-data, large-scale regime is impossible. Consequently, we restrict attention to a class of policies and focus on selecting a member that performs nearly as well as an oracle policy that knows the true value of μ in advance but is constrained to use a policy within the class. If the class of policies is sufficiently rich, we expect this oracle policy and, hence, our selected policy to have good practical performance in applications. Specifically, let $x(\theta, \hat{\mu})$ be a data-driven policy indexed by a parameter $\theta \in \Theta$, and let $\mathscr{X}^{\Theta}(\hat{\mu}) = \{x(\theta, \hat{\mu}) \in \mathscr{X} : \theta \in \Theta\}$ be a set of such policies.

Definition 3.1 (Oracle Policy)**.** Let

$$\theta^{\mathrm{OR}} = \theta^{\mathrm{OR}}(\mathrm{P}^{\mathrm{n}}, \hat{\mu}, \nu) \in \arg\max_{\theta \in \Theta} \frac{1}{n} \mu^{\top} x(\theta, \hat{\mu}).$$

We define the policy $\hat{\mu} \mapsto x(\theta^{\mathrm{OR}}, \hat{\mu})$ to be an *oracle policy* for problem P^{n} and policy class $\mathscr{X}^{\Theta}(\hat{\mu})$.

Note that θ^{OR} is random because it depends on $\hat{\mu}$. The performance of every policy in $\mathscr{X}^{\Theta}(\hat{\mu})$ is bounded above by the performance of $x(\theta^{\mathrm{OR}}, \hat{\mu})$ almost surely in $\hat{\mu}$; in this sense, the oracle policy serves as a benchmark. However, $x(\theta^{\mathrm{OR}}, \hat{\mu})$ is not a valid data-driven policy because identifying θ^{OR} requires knowing μ; in this sense, it is an "oracle." Indeed, it is not obvious that there exist data-driven policies that attain this benchmark without knowing μ.

We introduce two example policy classes that we study throughout the remainder of this paper.

Example 3.2 (Bayes-Inspired Policies)**.** In Section 4, we use a Bayesian argument to motivate the class of policies

$$\mathscr{X}^{Bayes}(\hat{\mu}) \equiv \{x(\tau, \hat{\mu}) \in \mathscr{X} : \tau \geq 0\},$$
$$\text{where } x(\tau, \hat{\mu}) \in \arg\max_{x \in \mathscr{X}} \frac{1}{n} \sum_{j=1}^{n} \frac{\nu_j}{\nu_j + \tau} \hat{\mu}_j x_j. \tag{3.1}$$

To avoid ambiguity, we index these policies by τ instead of θ. The oracle policy $x(\tau^{\mathrm{OR}}, \hat{\mu})$ is defined by $\tau^{\mathrm{OR}} \in \arg\max_{\tau \geq 0} \frac{1}{n} \sum_{j=1}^{n} \mu^{\top} x(\tau, \hat{\mu})$. We discuss properties of this oracle policy in Section 4.

Example 3.3 (Regularization Policies)**.** In Section 5, we motivate and study the class of policies

$$\mathscr{X}^{Reg}(\hat{\mu}) \equiv \{x^{\mathcal{R}}(\Gamma, \hat{\mu}) : \Gamma \in [\Gamma_{\min}, \Gamma_{\max}]\},$$
$$\text{where } x^{\mathcal{R}}(\Gamma, \hat{\mu}) \in \arg\max_{x \in \mathscr{X}} \frac{1}{n} \hat{\mu}^{\top} x - \frac{\Gamma \sqrt{\nu_{\min}}}{2n} \sum_{j=1}^{n} \frac{x_j^2}{\nu_j}. \tag{3.2}$$

Here $\Gamma_{\min}, \Gamma_{\max}$ are user-specified parameters. Again, to avoid ambiguity, we index this policy class by Γ instead of θ. Its oracle policy $x^{\mathcal{R}}(\Gamma^{\mathrm{OR}}, \hat{\mu})$ is defined by $\Gamma^{\mathrm{OR}} \in \arg\max_{\Gamma \in [\Gamma_{\min}, \Gamma_{\max}]} \frac{1}{n} \mu^{\top} x^{\mathcal{R}}(\Gamma, \hat{\mu})$.

Viewing P^{n} through the lens of a policy class also clarifies why SAA sometimes performs poorly in the small-data, large-scale regime.

Example 3.4 (SAA Revisited)**.** Suppose that $x^{SAA} \in \mathscr{X}^{\Theta}(\hat{\mu})$ a.s.; that is, there exists $\theta^{SAA} \in \Theta$ such that $x(\theta^{SAA}, \hat{\mu}) = x^{SAA}(\hat{\mu})$. Observe that $\theta^{SAA} \in \arg\max_{\theta \in \Theta} \frac{1}{n} \hat{\mu}^{\top} x(\theta, \hat{\mu})$ because

$$\begin{aligned} \hat{\mu}^{\top} x(\theta^{SAA}, \hat{\mu}) &= \hat{\mu}^{\top} x^{SAA}(\hat{\mu}) \\ &= \max_{x \in \mathscr{X}} \hat{\mu}^{\top} x \geq \sup_{\theta \in \Theta} \hat{\mu}^{\top} x(\theta, \hat{\mu}), \end{aligned}$$

where the last inequality follows because $x(\theta, \hat{\mu}) \in \mathscr{X}$ for all $\theta \in \Theta$.

Thus, SAA can be seen as finding the policy member that maximizes $\frac{1}{n} \hat{\mu}^{\top} x(\theta, \hat{\mu})$. By contrast, the oracle policy seeks to maximize the true reward $\frac{1}{n} \mu^{\top} x(\theta, \hat{\mu})$. Because of the dependence of $x(\theta, \hat{\mu})$ on $\hat{\mu}$, the SAA objective is biased. In the large-sample regime, this bias often vanishes asymptotically. However, in the small-data, large-scale regime, the bias is nonnegligible, partially explaining SAA's poor performance. We emphasize that if the policy class is large enough that $x^{SAA} \in \mathscr{X}^{\Theta}(\hat{\mu})$ a.s., its oracle policy necessarily performs at least as well as the SAA.

3.1. Our Bias-Corrected Policy

The main result of this section is a general framework for selecting a member of $\mathscr{X}^{\theta}(\hat{\mu})$ whose performance is comparable to $x(\theta^{\mathrm{OR}}, \hat{\mu})$ in the sense that we expect under mild conditions the suboptimality gap to be small in the small-data, large-scale regime. The key idea is to construct an approximation to $\mu^{\top} x(\theta, \hat{\mu})$ that does not require knowing μ and then to select $\hat{\theta}$ to optimize this approximation.

One such approximation might be $\hat{\mu}^{\top} x(\theta, \hat{\mu})$ because $\hat{\mu}$ is an unbiased estimator of μ. However, as noted in Example 3.4, this naive approximation is biased and should be corrected. Our bias correction is inspired by Stein's lemma (Ross et al. 2011), which states that for any (differentiable) function $f(\cdot)$ and $\zeta \sim \mathcal{N}(\mu, 1/\nu)$, we have $\mathbb{E}[(\zeta - \mu) f(\zeta)] = 1/\nu \cdot \mathbb{E}[\frac{\partial}{\partial \zeta} f(\zeta)]$. Considering the jth element of the bias $\mathbb{E}[(\hat{\mu} - \mu)^{\top} x(\theta, \hat{\mu}))]$, this roughly suggests that

$$\begin{aligned} \mathbb{E}[(\hat{\mu}_j - \mu_j) x_j(\theta, \hat{\mu})] &\approx \frac{1}{\nu_j} \mathbb{E}\left[\frac{\partial}{\partial \hat{\mu}_j} x_j(\theta, \hat{\mu})\right] \\ &\approx \frac{1}{2h\sqrt{\nu_j}} \mathbb{E}[x_j(\theta, \hat{\mu} + h e_j/\sqrt{\nu_j}) - x_j(\theta, \hat{\mu} - h e_j/\sqrt{\nu_j})], \end{aligned}$$

where e_j is the standard unit vector in the jth direction, and $0 < h < 1$ is a user-defined bandwidth parameter. For clarity, the last line approximates the derivative by a first-order finite difference with step $h/\sqrt{\nu_j}$, and the derivative in the second line may not be well defined because the solution $x(\theta, \hat{\mu})$ need not be differentiable (or even continuous). Lemma C.2 in Online Appendix C resolves these issues by generalizing Stein's lemma to approximately Gaussian random variables and approximate derivatives.

In any case, this heuristic argument suggests the bias-correction

$$B(\theta, h, \hat{\mu}) \equiv \frac{1}{n}\sum_{j=1}^{n}\frac{1}{2h\sqrt{\nu_j}}\left[x_j\left(\theta, \hat{\mu} + \frac{h}{\sqrt{\nu_j}}e_j\right) - x_j\left(\theta, \hat{\mu} - \frac{h}{\sqrt{\nu_j}}e_j\right)\right].$$

Approximating $\frac{1}{n}\mu^\top x(\theta, \hat{\mu})$ by $\frac{1}{n}\hat{\mu}^\top x(\theta, \hat{\mu}) - B(\theta, h, \hat{\mu})$, we let

$$\hat{\theta} \in \arg\max_{\theta\in\Theta}\frac{1}{n}\hat{\mu}^\top x(\theta, \hat{\mu}) - B(\theta, h, \hat{\mu}). \tag{3.3}$$

3.2. Performance in the Small-Data, Large-Scale Regime

The suboptimality of $x(\hat{\theta}, \hat{\mu})$ with respect to $x(\theta^{\mathsf{OR}}, \hat{\mu})$ depends on the quality of these approximations, which, in turn, depends on the policy class $\mathscr{X}^\Theta(\hat{\mu})$ and the estimator $\hat{\mu}$. We next prove a general-purpose bound that provides intuition into the types of policy classes and estimators for which this suboptimality gap is likely small in the small-data, large-scale regime. The key idea is that our bias-corrected criterion for $\hat{\theta}$ is very close to the true criterion that defines θ^{OR} in this regime.

Theorem 3.5 (A General Bound on the Suboptimality Gap). *Suppose that Assumption 2.1 holds and that there exists σ^2 such that for each j, the random variable $(\hat{\mu}_j - \mu_j)\sqrt{\nu_j}$ has a density $\phi_j(\cdot)$ and is sub-Gaussian with variance proxy at most σ^2.*[5] *Then*

$$\underbrace{\frac{1}{n}\mu^\top x(\theta^{\mathsf{OR}}, \hat{\mu}) - \frac{1}{n}\mu^\top x(\hat{\theta}, \hat{\mu})}_{\text{SubOptimality Gap}} \le \underbrace{\frac{4(h^{-1} + 24\sigma^2)}{\sqrt{\nu_{\min}}} \times \mathsf{TV} \times \log\left(\frac{e}{\mathsf{TV}}\right)}_{\text{Degree of NonNormality}} + \underbrace{\frac{4h^2}{\sqrt{\nu_{\min}}}}_{\text{Derivative Approximation}}$$
$$\underbrace{\begin{aligned}&+\sup_{\theta\in\Theta}\tfrac{1}{n}\left|(\hat{\mu}-\mu)^\top x(\theta, \hat{\mu}) - \mathbb{E}[(\hat{\mu}-\mu)^\top x(\theta, \hat{\mu})]\right| \\ &+\sup_{\theta\in\Theta}\left|B(\theta, h, \hat{\mu}) - \mathbb{E}[B(\theta, h, \hat{\mu})]\right|\end{aligned}}_{\text{Maximal Stochastic Deviations}},$$

where $\mathsf{TV} \equiv \frac{1}{2n}\sum_{j=1}^{n}\|\phi_j - \phi\|_1$ is a number between zero and one measuring the average total variation distance between $\phi_j(\cdot)$ and the standard normal density $\phi(\cdot)$.

Intuitively, Theorem 3.5 suggests that $x(\hat{\theta}, \hat{\mu})$ performs comparably to $x(\theta^{\mathsf{OR}}, \hat{\mu})$ in the small-data, large-scale regime whenever (1) the $\hat{\mu}_j$ are nearly Gaussian, (2) the $\hat{\mu}_j$ are sufficiently independent, and (3) $\mathscr{X}^\Theta(\hat{\mu})$ is not "too complex" a function class, as measured, for example, by its VC dimension.

To see this, note first that Theorem 3.5 decomposes the suboptimality gap into three terms. The first term is a deterministic constant measuring the degree of nonnormality in the $\hat{\mu}_j$ and notably does not depend on the instance P^n. It is small whenever TV is small. Moreover, when each $\hat{\mu}_j$ is Gaussian, then $(\hat{\mu}_j - \mu_j)\sqrt{\nu_j}$ is a standard normal random variable with $\phi_j(\cdot) = \phi(\cdot)$, so $\mathsf{TV} = 0$, and the first term on the right-hand side of the theorem vanishes.[6] The total variation TV thus measures the average degree of nonnormality of the $\hat{\mu}_j$. In particular, TV is small when each $\hat{\mu}_j$ is approximately Gaussian even if $\hat{\mu}$ is *not* jointly multivariate Gaussian.

The second term is also a deterministic constant stemming from our approximate Stein lemma (cf. Lemma C.2 in Online Appendix C) and can be made small through a suitable choice of bandwidth h. We discuss this choice in more detail later.

The third term, consisting of the two suprema, is more subtle and is a random variable. The argument of each suprema can be written as the average of n mean-zero random variables. Intuition suggests that for a fixed θ, these terms concentrate at zero (their mean) *provided that* the $\hat{\mu}_j$ are sufficiently independent as $n \to \infty$.[7] If, additionally, the policy class $\mathscr{X}^\Theta(\hat{\mu})$ is not too complex (i.e., it has low VC dimension or metric entropy), this concentration occurs uniformly for all $\theta \in \Theta$ (see, e.g., Pollard 1990, Van der Vaart 2000). Importantly, this convergence holds as $n \to \infty$ even if the ν_j remain bounded; that is, it holds in the small-data, large-scale regime. Commonly used policy classes in optimization, including those from Sections 4 and 5, often have low metric entropy, suggesting that this convergence holds for a wide variety of policy classes.

Stepping back, if the $\hat{\mu}_j$ are approximately Gaussian and sufficiently independent, then we expect, intuitively, that all three terms in the suboptimality bound are small in the small-data, large-scale regime for many policy classes. Making this intuition formal requires a careful analysis of the dependency in $x_j(\theta, \hat{\mu})$ across j and computing the "complexity" of $\mathscr{X}^\Theta(\hat{\mu})$. We carry out this analysis in Sections 4 and 5 for our Bayes-inspired and regularization policy classes in the special case of linear optimization problems with independent $\hat{\mu}_j$. This analysis confirms that these

policies have near-oracle performance in the small-data, large-scale regime in this setting.

Theorem 3.5 also highlights the trade-off in specifying the bandwidth h. As $h \to 0$, $\sup_{\theta\in\Theta} |B(\theta, h, \hat{\mu}) - \mathbb{E}[B(\theta, h, \hat{\mu})]|$ grows because $B(\theta, h, \hat{\mu})$ scales with $\frac{1}{h}$, and if $\mathsf{TV} \neq 0$, the *degree of nonnormality term* also grows. Thus, a good bandwidth must balance between these two terms and the error from derivative approximation in Theorem 3.5. One heuristic for selecting a bandwidth when $\mathsf{TV} = 0$ might be to select h large enough that $\theta \mapsto B(\theta, h, \hat{\mu})$ is smooth in θ because ideally the suprema is small and $\theta \mapsto \mathbb{E}[B(\theta, h, \hat{\mu})]$ is a smooth function of θ. Sections 4 and 5 provide more precise guidance on selecting a good bandwidth for those classes.

A potential drawback of our method (cf. Equation (3.3)) is that computing $B(\theta, h, \hat{\mu})$ for each θ seemingly involves solving $2n$ instances of P^{n} to determine $x(\theta, \hat{\mu} + \frac{h}{\sqrt{v_j}} e_j)$ and $x(\theta, \hat{\mu} - \frac{h}{\sqrt{v_j}} e_j)$ for each j (see, e.g., our example policy classes). For large n, this may be prohibitive. In Sections 4 and 5, we use the structure of the policy class to circumvent this issue, developing a more computationally efficient bias correction.

In summary, although Theorem 3.5 provides a framework for analyzing (cf. Equation (3.3)), at this level of generality, its usefulness is primarily foundational. It highlights the key conditions on the estimator and policy class required for good performance and also provides the road map for our more specialized analysis of specific policy classes (Sections 4 and 5). We expect that future research also might leverage Theorem 3.5 to study alternate optimization problems and policy classes by showing that the relevant maximal stochastic deviations are small.

3.3. Performance in the Large-Sample Regime

Before specializing and strengthening Theorem 3.5, we prove that despite being motivated by the small-data, large-scale regime, $\hat{\theta}$ has good theoretical performance in the large-sample regime. The key insight is that the bias correction is small in the large-sample regime. Thus, $x(\hat{\theta}, \hat{\mu})$ approximately optimizes the SAA objective, provided that $X_n^{\Theta}(\hat{\mu})$ contains $x^{SAA}(\hat{\mu})$ as a member.

Lemma 3.1 (Bound to SAA Performance). *Suppose that $x^{SAA}(\hat{\mu}) \in \mathscr{X}^{\Theta}(\hat{\mu})$ a.s. Then*

$$0 \le \frac{1}{n}\hat{\mu}^{\top}\left(x^{SAA}(\hat{\mu}) - x(\hat{\theta}, \hat{\mu})\right) \le \frac{1}{h\nu_{\min}},$$

$$\textit{where } \nu_{\min} = \min_{j=1,\ldots,n} \nu_j.$$

Under fairly general assumptions, SAA solutions perform comparably to the full-information optimum in the large-sample regime, that is, as $\nu_{\min}$ grows large. We use Lemma 3.1 to prove that our policy also performs well in these settings.

Theorem 3.6 (Bound to Full-Information Optimum). *Suppose that $x^{SAA}(\hat{\mu}) \in \mathscr{X}^{\Theta}(\hat{\mu})$ a.s. Then, under Assumption 2.1,*

$$\mathbb{E}\left[\left|Z^*(\mu) - \frac{1}{n}\mu^{\top}x(\hat{\theta}, \hat{\mu})\right|\right] \le \frac{2 + h^{-1}}{\sqrt{\nu_{\min}}}.$$

Theorem 3.6 is a nonasymptotic result but is perhaps best understood by considering a sequence of instances of $\{(\mathrm{P}^{\mathrm{n}}, \hat{\mu}^{\mathsf{S}}, \nu^{\mathsf{S}}) : \mathsf{S} \ge 1\}$ (indexed by S) in the large-sample regime. Because the bandwidth h scales with n, it is fixed in the large-sample regime, but $\nu_{\min}^{\mathsf{S}} \to \infty$ as $\mathsf{S} \to \infty$. Consequently, Theorem 3.6 proves that $x(\hat{\theta}, \hat{\mu})$ converges to full-information optimal performance in the large-sample regime even if the $\hat{\mu}$ are potentially highly nonnormal. We summarize these ideas in the following corollary, which proves that our policy also performs well in the large-sample regime.

Corollary 3.7 (Best of Both Regimes). *Consider a sequence $\{(P^n, \hat{\mu}^{\mathsf{S}}, \nu^{\mathsf{S}}) : \mathsf{S} \ge 1\}$ of instances in the large-sample regime, each satisfying Assumption 2.1. For each P^{n}, let $\hat{\theta}^{\mathsf{S}}$ be a solution to Equation (3.3). Assume for every S that $x^{SAA}(\hat{\mu}^{\mathsf{S}}) \in \mathscr{X}^{\Theta}(\hat{\mu}^{\mathsf{S}})$ a.s. Then, as $\mathsf{S} \to \infty$, the performance $\frac{1}{n}\mu^{\top}x(\hat{\theta}^{\mathsf{S}}, \hat{\mu}^{\mathsf{S}})$ converges in $\mathscr{L}_1$ to the full-information optimum Z^*.*

Corollary 3.7 provides a strong argument in favor of the policy $x(\hat{\theta}, \hat{\mu})$. Indeed, a potential criticism of our approach is that it limits attention to the class of policies $\mathscr{X}^{\Theta}(\hat{\mu})$, which may not contain the full-information optimum, although SAA (and similar approaches) is essentially nonparametric, considering all possible policies. Theorem 3.6 and Corollary 3.7 prove that despite restricting to a policy class, our proposed policy achieves performance comparable with the full-information optimum in the large-sample regime, provided that the policy class is rich enough to contain $x^{SAA}(\hat{\mu})$. We consider this a fairly minor condition, and it is satisfied, for example, by $\mathscr{X}^{Bayes}(\hat{\mu})$ and $\mathscr{X}^{Reg}(\hat{\mu})$ when $\Gamma_{\min} = 0$. Moreover, if, as in Example 1, $\nu_j = O(\mathsf{S})$ for all j, then the rate of convergence to full-information optimum is $O(1/\sqrt{\mathsf{S}})$, which is comparable with the convergence rate of the SAA solution (Kleywegt et al. 2002, Ruszczynski and Shapiro 2003).

4. Bayes-Inspired Policies over Polyhedral Feasible Regions

In this section, we specialize and strengthen Theorem 3.5 for the policy class in Example 3.2 with a polyhedral feasible region. We first motivate this policy class.

Consider a Bayesian approach to the decision problem $(\mathrm{P}^{\mathrm{n}}, \hat{\mu}, \nu)$. In this approach, we first assume that the unknown μ is drawn as a realization from a *known* prior distribution π; for example, we might assume $\mu_j \sim \mathcal{N}(0, 1/\tau_0)$ for some $\tau_0 \ge 0$ independently across j.

We then assume that the likelihood $\hat{\boldsymbol{\mu}} \mid \boldsymbol{\mu}$ follows a known likelihood distribution, for example, that $\hat{\mu}_j \mid \mu_j \sim \mathcal{N}(\mu_j, 1/\nu_j)$ independently across j. In principle, we can then compute the Bayes-optimal policy with respect to π using the data $\hat{\boldsymbol{\mu}}$. For Gaussian priors and likelihoods, conjugacy gives the posterior mean $\mathbb{E}^{\pi}[\mu_j \mid \hat{\boldsymbol{\mu}}] = \frac{\nu_j}{\nu_j+\tau_0}\hat{\mu}_j$, and it follows that $x(\tau_0, \hat{\boldsymbol{\mu}})$ defined in Example 3.2 is a Bayes-optimal policy.

Bayes' policies enjoy strong performance guarantees. For example, Bayes' policies are always admissible; that is, no other data-driven policy Pareto-dominates a Bayes optimal policy (Berger 2013). Under mild assumptions, Bayes' optimal policies also form an essentially complete class; that is, any data-driven policy is weakly dominated by a Bayes optimal policy for some prior (Berger 2013).

Nonetheless, Bayes' policies require strong assumptions, for example, that one knows the precise form of the prior. We can partially mitigate these drawbacks by not fixing a prior and instead considering a class of policies, each of which is Bayes optimal for some prior, and then seeking its oracle member. Indeed, this perspective motivates our policy class $\mathcal{X}^{Bayes}(\hat{\boldsymbol{\mu}})$; that is, it is the set of Bayes-optimal policies when the μ_j are drawn from some mean-zero Gaussian prior, and the $\hat{\mu}_j \mid \mu_j$ are Gaussian, justifying our terminology *Bayes inspired*.[8]

We argue that even if one does not adopt the Bayesian perspective (i.e., one does *not* believe that $\boldsymbol{\mu}$ is drawn from some mean-zero Gaussian prior), $\mathcal{X}^{Bayes}(\hat{\boldsymbol{\mu}})$ is still a rich and interesting class of policies for two reasons. First, $\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}})$ has an intuitive structure. Each estimate $\hat{\mu}_j$ is shrunk toward zero. Components with lower precision are shrunk more aggressively because if $\nu_j < \nu_k$, then $\frac{\nu_j}{\nu_j+\tau} \le \frac{\nu_k}{\nu_k+\tau}$. The parameter τ controls the degree of shrinkage.

Second, a host of popular estimators for $\boldsymbol{\mu}$ have the general form $(\frac{\nu_j}{\nu_j+\tau}\hat{\mu}_j \;:\; j = 1, \ldots, n)$ for some data-driven value of τ when $\hat{\boldsymbol{\mu}}$ is Gaussian. These estimators often have provably good theoretical properties and exhibit good empirical performance in applications even when the underlying Gaussian assumptions might be violated. The class $\mathcal{X}^{Bayes}(\hat{\boldsymbol{\mu}})$ thus contains the corresponding estimate-then-optimize policy for each of these estimators.[9] Consequently, $\boldsymbol{x}(\tau^{\mathrm{OR}}, \hat{\boldsymbol{\mu}})$ must perform at least as well as these estimate-then-optimize policies. We consider this to be a strong argument for good performance of the oracle policy in a broad range of applications. Examples of such estimators and their properties when $\hat{\boldsymbol{\mu}}$ is Gaussian include

- *Maximum likelihood estimation.* The value $\tau = 0$ yields the estimator $\hat{\boldsymbol{\mu}}$, which is the maximum likelihood estimator for $\boldsymbol{\mu}$ and also the minimum variance unbiased estimator. Moreover, by construction, $\boldsymbol{x}(0, \hat{\boldsymbol{\mu}})$ equals the sample average approximation $\boldsymbol{x}^{SAA}(\hat{\boldsymbol{\mu}})$ because $\boldsymbol{x}(0, \hat{\boldsymbol{\mu}}) = \arg\max_{\boldsymbol{x}\in\mathcal{X}} \frac{1}{n}\hat{\boldsymbol{\mu}}^{\top}\boldsymbol{x}$. Because the SAA policy is a member of $\mathcal{X}^{Bayes}(\hat{\boldsymbol{\mu}})$, it follows that its performance must be no better than that of the oracle policy. In fact, as shown in Example 4.1, its performance can be much worse.
- *James–Stein shrinkage.* The value $\tau^{JS} = 1 - (1 - \frac{n-2}{\|\hat{\boldsymbol{\mu}}\|_2^2})^{-1}$ yields the James–Stein estimator. When $\nu_j = 1$ for all $j = 1, \ldots, n$ and $n \ge 3$, Stein (1981) proved that this biased estimate has smaller mean-squared error than $\hat{\boldsymbol{\mu}}$.
- *Empirical Bayes.* The values τ^{MM} and τ^{MLE}, defined by

$$\tau^{MM} \equiv \left(\frac{1}{n}\sum_{j=1}^{n}\left(\hat{\mu}_j^2 - 1/\nu_j\right)\right)^{-1} \quad \text{and}$$

$$\tau^{MLE} \text{ is a solution to } \sum_{j=1}^{n}\frac{\hat{\mu}_j^2 - 1/\tau^{MLE} - 1/\nu_j}{\left(1/\tau^{MLE} + 1/\nu_j\right)^2} = 0,$$

correspond, respectively, to the empirical Bayes moment-matching and empirical Bayes maximum likelihood estimators (Xie et al. 2012).[10] Generally speaking, parametric empirical Bayes methods assume that the data were generated from a Bayesian hierarchical model and then use the marginal distribution of the data to fit parameters of the prior. For our specific case, this amounts to assuming that μ_j is a realization of a mean-zero Gaussian prior and then using either the method of moments or maximum likelihood estimation on $\hat{\boldsymbol{\mu}}$ to estimate τ (see Xie et al. (2012) for details). If the assumed hierarchical model is actually valid, the two methods both converge in performance as $n \to \infty$ to the Bayes optimal procedure.
- *SURE Estimation.* Finally, Xie et al. (2012) propose the choice τ^{SURE}, which solves[11]

$$\sum_{j=1}^{n}\left(\frac{\nu_j^{-2}\hat{\mu}_j^2}{\left(1/\nu_j + 1/\tau^{SURE}\right)^3} - \frac{\nu_j^{-2}}{\left(1/\nu_j + 1/\tau^{SURE}\right)^2}\right) = 0.$$

Xie et al. (2012) prove that as $n \to \infty$, this estimator achieves the minimum mean-squared error among all estimators of the form $(\frac{\nu_j}{\nu_j+\tau}\hat{\mu}_j \;:\; j = 1, \ldots, n)$ a.s.

By construction, $\boldsymbol{x}(\tau^{\mathrm{OR}}, \hat{\boldsymbol{\mu}})$ is no worse than the preceding estimate-then-optimize policies *whether* $\boldsymbol{\mu}$ *is a realization from a Gaussian prior*.

A potential criticism of estimate-then-optimize policies is that the estimation phase involves a purely statistical criterion that is agnostic to the downstream optimization (Liyanage and Shanthikumar 2005, Elmachtoub and Grigas 2017). By contrast, $\boldsymbol{x}(\tau^{\mathrm{OR}}, \hat{\boldsymbol{\mu}})$ does leverage optimization structure. Consequently, as shown in Example 4.1, it may perform strictly better than these estimate-then-optimize policies.

Example 4.1 (Benefits over Estimate-then-Optimize). Consider the ranking problem from Example 3 with $\alpha = 0.05$, $\hat{\mu}_j \sim \mathcal{N}(\mu_j, 1/\nu_j)$ independently across j, and

$$(\mu_j, \nu_j) = \begin{cases} (0.0,\ 0.1), & \text{if} \quad 1 \le j \le \lfloor n/3 \rfloor, \\ (0.3,\ 4.0), & \text{if} \ \lfloor n/3 \rfloor < j \le \lfloor 2n/3 \rfloor, \\ (1.0,\ 1.0), & \text{if} \ \lfloor 2n/3 \rfloor < j \le n. \end{cases}$$

We have three types of items in roughly equal proportions. *Low* items have low value ($\mu_j = 0$) and low precision ($\nu_j = 0.1$), so their noisy estimates frequently make them appear attractive. *Medium* items have medium value and very high precision ($\mu_j = 0.3$ and $\nu_j = 4$). *High* items have good value and medium precision ($\mu_j = 1$ and $\nu_j = 1$). Note that these values of μ_j do not resemble a draw from a Gaussian prior and that a good solution contains many high items.

The left panel of Figure 2 shows the performance of $\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}})$ for varying τ, a single realization of $\hat{\boldsymbol{\mu}}$, and $n = 512$ and $n = 2^{17}$. The right panel shows the performance of a subset of these estimate-then-optimize policies $\boldsymbol{x}(\tau^{\mathrm{OR}}, \hat{\boldsymbol{\mu}})$ and our proposed policy $\boldsymbol{x}(\hat{\tau}, \hat{\boldsymbol{\mu}})$ (defined in the next section). See Online Appendix F for a larger figure including all the estimate-then-optimize policies.

The example highlights the trade-offs induced by τ. Note that for any τ, $\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}})$ corresponds to ranking the rewards $\frac{\nu_j}{\nu_j + \tau}\hat{\mu}_j$ and selecting the top 5% of items. When $\tau = 0$, we have no shrinkage, the low items appear very attractive, and $\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}})$ chooses many low items. This is illustrated in Figure 3(a), which shows the distribution of the reward and that low items comprise the tail of the distribution. If τ is too large, such as $\tau = 4.94$, there is "too much" shrinkage, and the medium items comprise the tail of the distribution, as shown in Figure 3(c). For intermediate values of τ, say $\tau = 0.577$, high items appear the most attractive (cf. Figure 3(b)).

This trade-off, in turn, drives the performance of the various methods. Indeed, in Figure 3(d), we see that, for this example, τ^{OR} is fairly small but strictly positive. Hence, SAA ($\tau = 0$) shrinks too little, and the estimate-then-optimize policies generally shrink too much. More specifically, the estimate-then-optimize policies seek τ values that yield low mean-squared error, a statistical objective. Because of the form of our optimization problem, however, lower mean-squared error does not translate into better performance.

By contrast, we observe that our proposed policy $\boldsymbol{x}(\hat{\tau}, \hat{\boldsymbol{\mu}})$ (defined in the next section) converges in performance to $\boldsymbol{x}(\tau^{\mathrm{OR}}, \hat{\boldsymbol{\mu}})$ as $n \to \infty$. As we see in the next section, this policy *does* explicitly leverage the optimization structure.

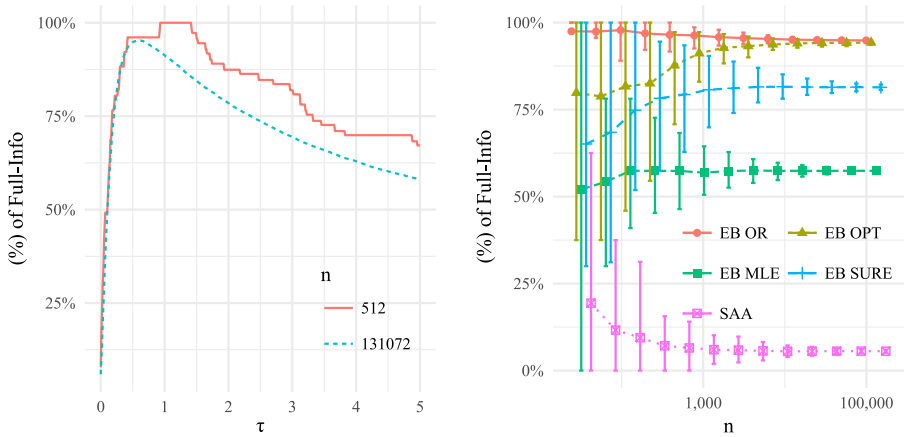
4.1. A Near-Oracle Policy for the Bayes-Inspired Class

Instead of applying Theorem 3.5 verbatim to bound the suboptimality, we use the special structure of $\mathcal{X}^{Bayes}(\hat{\boldsymbol{\mu}})$ to construct a more computationally efficient bias correction and focus on a special case of P^{n} to develop further insight. Specifically, we assume that $\mathcal{X}$ is polyhedral, that is,

$$\mathcal{X} = \left\{ \boldsymbol{x} \in [0,1]^n : \frac{1}{n}\sum_{j=1}^{n} \boldsymbol{A}_j x_j \le \boldsymbol{b} \right\}, \tag{4.1}$$

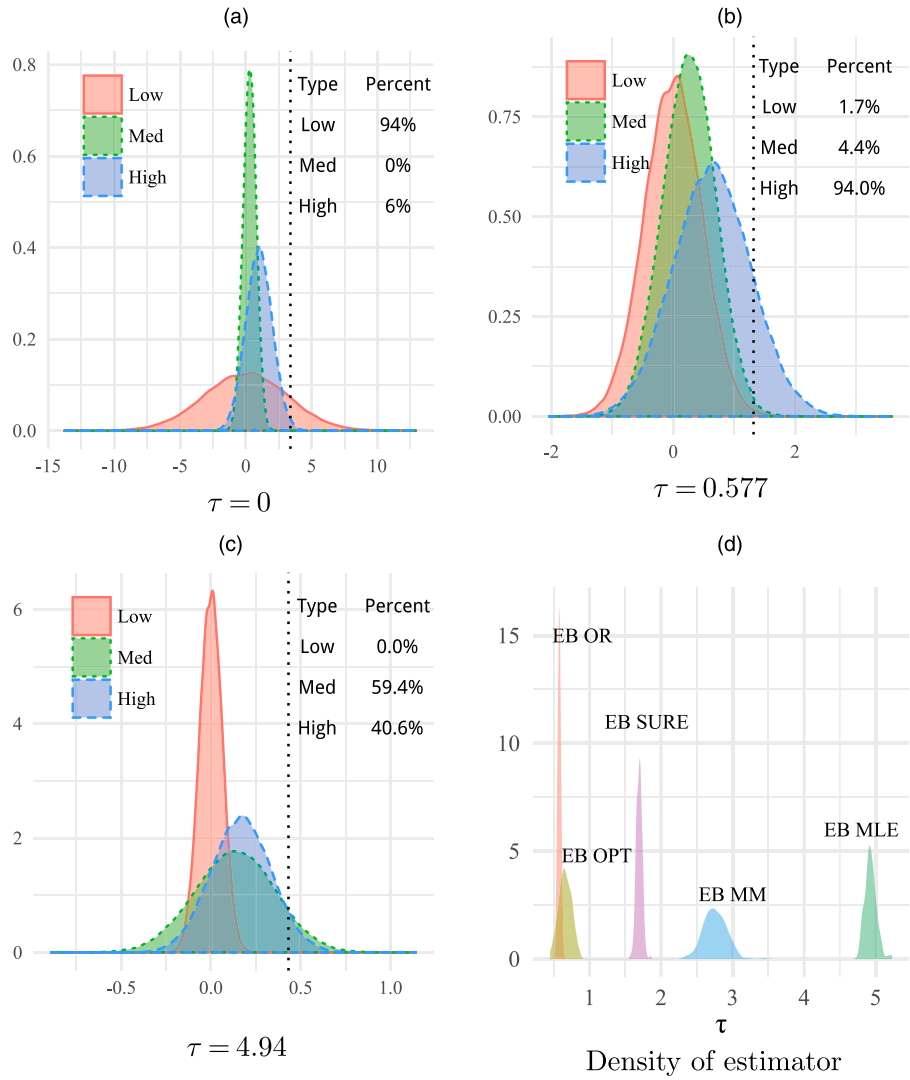
where $\boldsymbol{b} \in \mathbb{R}^m$ is an arbitrary vector, so that P^{n} represents an arbitrary linear optimization problem over a bounded feasible region. Linear optimization is a

Figure 2. (Color online) Benefits over Estimate-then-Optimize in Example 4.1



Notes. The left panel shows the performance of $\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}})$ for varying τ, $n = 2^{17} = 131{,}072$ (dotted line) and $n = 2^9 = 512$ (solid line). The function is discontinuous but becomes smoother as $n \to \infty$. The right panel shows the performance of various data-driven policies from $\mathcal{X}^{Bayes}(\hat{\boldsymbol{\mu}})$ for varying n: SAA, sample average approximation; EB OR, empirical Bayes oracle benchmark; EB MLE, empirical Bayes maximum likelihood estimate; EB SURE, empirical Bayes Stein's unbiased risk estimate; EB OPT, (defined in Section 4) our proposed procedure. Only EB OPT achieves best-in-class performance.

Figure 3. (Color online) Effects of Shrinkage on Solution in Example 4.1



Notes. Parts (a)–(c) show smoothed histograms of $v_j\hat{\mu}_j/(v_j+\tau)$ for various τ along a single sample path, $n = 2^{17} = 131,072$. When $\alpha = 0.05$, all items to the right of the dotted black line are chosen. Although a small amount of shrinkage increases the number of high items chosen, excessive shrinkage causes many medium items to be eventually chosen. Part (d) shows the density of the fitted τ for various methods over 200 simulations. Estimate-then-optimize methods generally overshrink for this example.

fundamental class of problems subsuming many applications. Any linear optimization with a bounded feasible region can be written in the form of problem P^{n} after potentially scaling and shifting to place the feasible region in the unit box $[0,1]^n$. In particular, for a fixed n, scaling the constraints by $1/n$ is without loss of generality, and our results are stated for fixed n.

We stress that $\mathscr{X}$ and hence $\boldsymbol{A}$ and $\boldsymbol{b}$ are known a priori. However, with no further assumptions, $\mathscr{X}$ may be empty, rendering P^{n} trivial. To avoid technicalities, we assume that P^{n} is strictly feasible.

Definition 4.1 (s_0-Strict Feasibility)**.** We say that $\mathscr{X}$ is *s_0-strictly feasible* if there exists $s_0 > 0$ and $\boldsymbol{x}_0 \in \mathscr{X}$ such that $\frac{1}{n}\sum_{j=1}^{n} \boldsymbol{A}_j x_j^0 + s_0\boldsymbol{e} \le \boldsymbol{b}$.

Assuming strict feasibility is slightly stronger than assuming that $\mathscr{X}$ is nonempty, but it can often be achieved via a small perturbation of $\boldsymbol{A}$ and $\boldsymbol{b}$.[12]

To define our specialized policy, we first reinterpret $\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}})$ as the solution to

$$\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}}) \in \arg\max_{\boldsymbol{x}\in\mathscr{X}} \frac{1}{n}\sum_{j=1}^{n} r_j(\tau, \hat{\mu}_j) x_j,$$
$$\text{where} \quad r_j(\tau, \hat{\mu}_j) = \frac{v_{\min}+\tau}{v_{\min}} \times \frac{v_j}{v_j+\tau}\hat{\mu}_j. \tag{4.2}$$

The objective coefficients in Equations (3.1) and (4.2) differ by a positive scaling, so the two definitions of $\boldsymbol{x}(\tau, \hat{\boldsymbol{\mu}})$ are equivalent. However, an advantage of Equation (4.2) is that the variance of $r_j(\tau, \hat{\mu}_j)$ is bounded below by $1/v_j$, which simplifies the analysis.

Under Equation (4.1), the dual to Equation (4.2) is

$$\min_{\boldsymbol{\lambda}\in\mathbb{R}^m_+} D_{\hat{\boldsymbol{\mu}}}(\boldsymbol{\lambda},\tau),$$

$$\text{where}\quad D_{\hat{\boldsymbol{\mu}}}(\boldsymbol{\lambda},\tau) \equiv \boldsymbol{b}^\top\boldsymbol{\lambda} + \frac{1}{n}\sum_{j=1}^{n}\left(r_j(\tau,\hat{\mu}_j) - \boldsymbol{A}_j^\top\boldsymbol{\lambda}\right)^+.$$

Let $\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})$ be an optimal solution. Now we can define our alternate bias correction: for any bandwidth $0<h<1$, define $h_j(\tau) = \frac{\nu_{\min}+\tau}{\nu_{\min}}\times\frac{h\sqrt{\nu_j}}{\nu_j+\tau} = r_j(\tau, h/\sqrt{\nu_j})$, and let

$$B^{Bayes}(\tau,h,\hat{\boldsymbol{\mu}}) \equiv \frac{1}{n}\sum_{j=1}^{n}\frac{1}{2h\sqrt{\nu_j}}\mathbb{I}\left(\left|r_j(\tau,\hat{\mu}_j) - \boldsymbol{A}_j^\top\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})\right| \le h_j(\tau)\right),$$

and define our candidate policy $\boldsymbol{x}(\hat{\tau},\hat{\boldsymbol{\mu}})$ by

$$\hat{\tau} \in \arg\max_{\tau\ge 0}\ \frac{1}{n}\hat{\boldsymbol{\mu}}^\top\boldsymbol{x}(\tau,\hat{\boldsymbol{\mu}}) - B^{Bayes}(\tau,h,\hat{\boldsymbol{\mu}}). \tag{4.3}$$

Note that computing $\hat{\tau}$ does *not* require knowledge of $\boldsymbol{\mu}$. We motivate this particular form of the bias correction in the next section. Note that we can compute $\hat{\tau}$ using a grid search.

We next bound the suboptimality $\boldsymbol{x}(\hat{\tau},\hat{\boldsymbol{\mu}})$ against the oracle for $\mathscr{X}^{Bayes}(\hat{\boldsymbol{\mu}})$. To develop explicit tail bounds on the stochastic terms for all finite n, we require the following assumptions.

Assumption 4.2 (Independent Near-Gaussians). *In addition to Assumption 2.1, assume that*

a. *(Independence) The random variables $\hat{\mu}_1,\ldots,\hat{\mu}_n$ are independent.*

b. *(Bounded precisions) There exists $\nu_{\min},\nu_{\max}$ such that $0<\nu_{\min}\le\nu_j\le\nu_{\max}<\infty$ for all $j=1,\ldots,n$.*

c. *(Sub-Gaussian) There exists a positive constant σ such that for all $j=1,\ldots,n$, $(\hat{\mu}_j-\mu_j)\sqrt{\nu_j}$ is sub-Gaussian with variance proxy at most σ^2.*

d. *(Bounded and positive density) For all $j=1,\ldots,n$, $(\hat{\mu}_j-\mu_j)\sqrt{\nu_j}$ admits a density $\phi_j(\cdot)$ that is bounded and strictly positive over any finite interval; that is, $\max_{j=1,\ldots,n}\sup_{t\in\mathbb{R}}\phi_j(t)<\infty$ and $\min_{j=1,\ldots,n}\inf_{t:|t|\le T}\phi_j(t)>0$ for all $T>0$.*

These assumptions impose rather mild requirements on the distribution of $\hat{\mu}_j$. When the $\hat{\mu}_j$ are Gaussian, then $\phi_j(\cdot)$ is the density of the standard Gaussian, which is bounded above by $1/\sqrt{2\pi}$ and below by $\frac{1}{\sqrt{2\pi}}\exp(-\frac{T^2}{2})$ over an interval $[-T,T]$. Intuitively, if the $\hat{\mu}_j$ are approximately normally distributed, we expect these conditions to also hold. We can now strengthen Theorem 3.5 for $\boldsymbol{x}(\tau,\hat{\boldsymbol{\mu}})$ by bounding the maximal stochastic deviations.

Theorem 4.3 (Finite-Dimension Bound for the Bayes-Inspired Policy Class). *Consider P^n under Assumption 4.2 with $\mathscr{X}$ given by Equation (4.1) such that $\mathscr{X}$ is s_0-strictly feasible and $m\ge 1$. Let C_μ, C_A be such that $|\mu_j|\le C_\mu$ and $\|\boldsymbol{A}_j\|_\infty\le C_A$ for all $j=1,\ldots,n$ and $\beta>0$ be such that the minimum eigenvalue of the matrix $\frac{1}{n}\sum_{j=1}^n\boldsymbol{A}_j\boldsymbol{A}_j^\top\in\mathbb{R}^{m\times m}$ is at least β. Then, for all $0<\delta<1$, $0<h<1$, there exist positive constants C_1,C_2 not depending on $\{n,m,\delta,h\}$ such that*

$$0 \le \frac{1}{n}\boldsymbol{\mu}^\top\left(\boldsymbol{x}(\tau^{\mathrm{OR}},\hat{\boldsymbol{\mu}}) - \boldsymbol{x}(\hat{\tau},\hat{\boldsymbol{\mu}})\right)$$
$$\le 2C_1\left(\underbrace{\frac{\mathsf{TV}\log(e/\mathsf{TV})}{h}}_{\text{Degree of NonNormality}} + \underbrace{\frac{\delta}{h}}_{\substack{\text{Approximating}\\ \text{Dual Solution}}} + \underbrace{h^2}_{\substack{\text{Approximating}\\ \text{Stein's Lemma}}}\right) + \underbrace{2R}_{\substack{\text{Stochastic}\\ \text{Errors}}},$$

where, for any $\epsilon > \frac{m^2\log n}{n^{3/2}\log(m+1)}$,

$$\mathbb{P}\{R>6\epsilon\} \le 130\left[\exp\left(\frac{-C_2\delta\sqrt{n}}{\sqrt{m}\log(m+1)}\right)\times\log\left(1+\frac{\sqrt{m}\log(m+1)}{C_2\delta\sqrt{n}}\right) + \exp\left(\frac{-C_2\epsilon h\sqrt{n}}{\log(m+1)}\right)\right],$$

and $\mathsf{TV} = \frac{1}{2n}\sum_{j=1}^n\|\phi_j-\phi\|_1$ is the average total variation distance between $\phi_j(\cdot)$ and the standard normal density $\phi(\cdot)$.

Theorem 4.3 decomposes the suboptimality gap into a deterministic error arising from various approximation steps in our proof and a stochastic error with bounded tails. The result is not asymptotic; it holds for a fixed instance of $(\mathrm{P}^n,\hat{\boldsymbol{\mu}},\boldsymbol{\nu})$, that is, for finite n. Similar to Theorem 3.5, the bound suggests that our method performs well in the small-data, large-scale regime if the $\hat{\mu}_j$ are nearly Gaussian. Specifically, for a well-chosen sequence of bandwidths h_n depending on n, the performance of our policy converges to that of the oracle asymptotically, as in the following corollary, whose proof is given in Section D.3 in Online Appendix D.

Corollary 4.4 (Almost-Sure Convergence to Oracle). *Consider a sequence $\{(P^n,\hat{\boldsymbol{\mu}}^n,\boldsymbol{\nu}^n): n\ge 2\}$ of instances in the small-data, large-scale regime, each satisfying the hypothesis of Theorem 4.3 such that the parameters do not grow with n; that is, m is constant, $\|\boldsymbol{A}_j\|\le C_A$, $|\mu_j^n|\le C_\mu$, and $\nu_{\min}\le\nu_j^n\le\nu_{\max}$ for all $1\le j\le n$. Suppose further that the smallest eigenvalue of $\frac{1}{n}\sum_{j=1}^n\boldsymbol{A}_j^n\boldsymbol{A}_j^{n\top}\in\mathbb{R}^{m\times m}$ is at least β for all n. Let h_n be bandwidth parameters chosen such that $h_n\to 0$ and $h_n\sqrt{n}\to\infty$. Finally, for each P^n, let $\hat{\tau}^n$ be the solution to Equation (4.3) with bandwidth h_n. If each $\hat{\mu}_j$ is Gaussian, then the policy $\boldsymbol{x}(\hat{\tau}^n,\hat{\boldsymbol{\mu}})$ performs as well as the oracle policy for $\mathscr{X}^{Bayes,n}(\hat{\boldsymbol{\mu}})$ a.s. as $n\to\infty$.*

For the Gaussian case with $\mathsf{TV}=0$, by matching the orders of the deterministic and stochastic errors in Theorem 4.3, one can show that the rate of convergence for the bound is optimized when $h_n = O(n^{-1/6})$, in which case the suboptimality gap converges to zero

as $O_p(n^{-1/3})$. Similar arguments hold when some parameters grow mildly with n, such as $m = O(\log n)$, with slightly modified bandwidth sequences. Regardless of the precise scaling, Theorem 4.3 provides good evidence that our procedure based on $\hat{\tau}$ should have a strong performance for instances with large, finite n as long as the other parameters are not too large. We confirm this claim numerically in Section 6.

In the remainder of this section, we motivate our alternate bias correction in Equation (4.3) and outline the proof of Theorem 4.3 by considering the asymptotic regime in which $n \to \infty$ while other parameters stay fixed. See Section D.1 in Online Appendix D for a formal proof for finite n, including explicit values for the constants C_1 and C_2.

4.2. Proof Outline for Theorem 4.3 and the Alternate Bias Correction

The proof of Theorem 4.3 follows the framework of that of Theorem 3.5; that is, we first seek to prove that

$$\sup_{\tau\geq 0}\frac{1}{n}\left|\sum_{j=1}^{n}\left(\hat{\mu}_j-\mu_j\right)x_j(\tau,\hat{\mu})-\mathbb{E}\left[\left(\hat{\mu}_j-\mu_j\right)x_j(\tau,\hat{\mu})\right]\right|\to_p 0. \tag{4.4}$$

Establishing this convergence involves two key ideas.

Rounding the Primal Solution (Lemma D.6 in Online Appendix D). We first use linear optimization duality to rewrite $\boldsymbol{x}(\tau,\hat{\boldsymbol{\mu}})$ more simply. By complementary slackness, for all but possibly m components in which the x_j are fractional, we have that

$$x_j(\tau,\hat{\mu}_j)=\mathbb{I}\left(r_j(\tau,\hat{\mu}_j)>\boldsymbol{A}_j^\top\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})\right). \tag{4.5}$$

These fractional terms contribute at most $\frac{m}{n}\max_{j=1,\ldots,n}|\mu_j-\hat{\mu}_j|$ to the total. Because the maximum of sub-Gaussian random variables concentrates at its mean, which grows like $O(\sqrt{\log n})$, the error from rounding the primal solution is small.

$$\frac{1}{n}(\boldsymbol{\mu}-\hat{\boldsymbol{\mu}})^\top\boldsymbol{x}(\tau,\hat{\boldsymbol{\mu}})=\frac{1}{n}\sum_{j=1}^{n}(\mu_j-\hat{\mu}_j)\mathbb{I}\left(r_j(\tau,\hat{\mu}_j)>\boldsymbol{A}_j^\top\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})\right)+o_p(1). \tag{4.6}$$

Approximating the Dual Solution (Lemma D.7 in Online Appendix D). The sum in Equation (4.6) consists of *dependent* random variables because $\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})$ depends on the entire vector $\hat{\boldsymbol{\mu}}$. This dependence poses a technical challenge in establishing the requisite convergence. The second key idea approximates $\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})$ by the solution of an "average" dual problem to break this dependence. Define

$$\min_{\boldsymbol{\lambda}\in\mathbb{R}_+^m} D(\boldsymbol{\lambda},\tau),$$
$$\text{where } D(\boldsymbol{\lambda},\tau)\equiv\boldsymbol{b}^\top\boldsymbol{\lambda}+\frac{1}{n}\sum_{j=1}^{n}\mathbb{E}\left[\left(r_j(\tau,\hat{\mu}_j)-\boldsymbol{A}_j^\top\boldsymbol{\lambda}\right)^+\right],$$

and let $\boldsymbol{\lambda}(\tau)$ be an optimal solution. Intuitively, $\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})=\boldsymbol{\lambda}(\tau)+o_p(1)$ for large n (Lemma D.5 in Online Appendix D) so that the error from approximating the dual solution is also small.

$$\frac{1}{n}(\boldsymbol{\mu}-\hat{\boldsymbol{\mu}})^\top\boldsymbol{x}(\tau,\hat{\boldsymbol{\mu}})=\frac{1}{n}\sum_{j=1}^{n}(\mu_j-\hat{\mu}_j)\mathbb{I}\left(r_j(\tau,\hat{\mu}_j)>\boldsymbol{A}_j^\top\boldsymbol{\lambda}(\tau)\right)+o_p(1).$$

Importantly, replacing $\boldsymbol{\lambda}(\tau,\hat{\boldsymbol{\mu}})$ with $\boldsymbol{\lambda}(\tau)$ transforms the sum into a sum of *independent* random variables. We can now use standard uniform law of large number results (Lemma D.8 in Online Appendix D) to establish

$$\frac{1}{n}(\boldsymbol{\mu}-\hat{\boldsymbol{\mu}})^\top\boldsymbol{x}(\tau,\hat{\boldsymbol{\mu}})=\frac{1}{n}\sum_{j=1}^{n}\mathbb{E}\left[(\mu_j-\hat{\mu}_j)\mathbb{I}\left(r_j(\tau,\hat{\mu}_j)>\boldsymbol{A}_j^\top\boldsymbol{\lambda}(\tau)\right)\right]+o_p(1). \tag{4.7}$$

Finally, by "unwinding" these approximations, we can also show that

$$\frac{1}{n}\sum_{j=1}^{n}\mathbb{E}\left[(\mu_j-\hat{\mu}_j)\mathbb{I}\left(r_j(\tau,\hat{\mu}_j)>\boldsymbol{A}_j^\top\boldsymbol{\lambda}(\tau)\right)\right]=\frac{1}{n}\sum_{j=1}^{n}\mathbb{E}\left[(\hat{\mu}_j-\mu_j)x_j(\tau,\hat{\boldsymbol{\mu}})\right]+o_p(1),$$

which proves Equation (4.4).

If we sought to apply Theorem 3.5 directly, we would next need to prove

$$\frac{1}{n}\sum_{j=1}^{n}\frac{1}{2h\sqrt{v_j}}\left[x_j\left(\tau,\hat{\boldsymbol{\mu}}+\frac{h}{\sqrt{v_j}}\boldsymbol{e}_j\right)-x_j\left(\tau,\hat{\boldsymbol{\mu}}-\frac{h}{\sqrt{v_j}}\boldsymbol{e}_j\right)\right]\to_p 0.$$

This convergence can be established using similar arguments employing the average dual optimal solution, but it is easier to take a different approach. Specifically, instead of applying our approximate Stein's lemma to $x_j(\cdot)$ as in Theorem 3.5, we apply it to the right-hand side of Equation (4.7). This leads to the third key idea in the proof.

A Custom Bias Correction. Let $\zeta_j = (\hat{\mu}_j - \mu_j)\sqrt{\nu_j}$, and let $f_j : \mathbb{R} \to \mathbb{R}$ be defined by $f_j(t) = \frac{-1}{\sqrt{\nu_j}}\mathbb{I}(r_j(\tau, \mu_j + t/\sqrt{\nu_j}) > bA_j^\top \lambda(\tau))$ for all $t \in \mathbb{R}$. Note that ζ_j is mean zero, has precision one, and is sub-Gaussian with parameter σ^2. Furthermore,

$$\mathbb{E}\Big[(\mu_j - \hat{\mu}_j)\mathbb{I}\Big(r_j(\tau, \hat{\mu}_j) > A_j^\top \lambda(\tau)\Big)\Big] = \mathbb{E}[\zeta_j f_j(\zeta_j)].$$

Using Lemma C.2 in Online Appendix C, we approximate $\mathbb{E}[\zeta_j f_j(\zeta_j)]$ by a first-order finite difference with step h, so

$$\begin{aligned}
&\mathbb{E}[\zeta_j f_j(\zeta_j)] \\
&\quad \approx \frac{1}{2h}\mathbb{E}[f_j(\zeta_j + h) - f_j(\zeta_j - h)] \\
&\quad = \frac{-\begin{pmatrix} \mathbb{I}\Big\{r_j\Big(\tau, \hat{\mu}_j + h/\sqrt{\nu_j}\Big) > A_j^\top \lambda(\tau)\Big\} \\ -\mathbb{I}\Big\{r_j\Big(\tau, \hat{\mu}_j - h/\sqrt{\nu_j}\Big) > A_j^\top \lambda(\tau)\Big\} \end{pmatrix}}{2h\sqrt{\nu_j}}.
\end{aligned}$$

Because $\hat{\mu}_j \mapsto r_j(\tau, \hat{\mu}_j)$ is linear, $r_j(\tau, \hat{\mu}_j + h/\sqrt{\nu_j}) = r_j(\tau, \hat{\mu}_j) + r_j(\tau, h/\sqrt{\nu_j}) = r_j(\tau, \hat{\mu}_j) + h_j(\tau)$, and similarly, $r_j(\tau, \hat{\mu}_j - h/\sqrt{\nu_j}) = r_j(\tau, \hat{\mu}_j) - h_j(\tau)$. Thus,

$$\begin{aligned}
&\mathbb{I}\Big\{r_j\Big(\tau, \hat{\mu}_j + h/\sqrt{\nu_j}\Big) > A_j^\top \lambda(\tau)\Big\} \\
&\quad - \mathbb{I}\Big\{r_j\Big(\tau, \hat{\mu}_j - h/\sqrt{\nu_j}\Big) > A_j^\top \lambda(\tau)\Big\} \\
&\quad = \mathbb{I}\Big\{r_j(\tau, \hat{\mu}_j) + h_j\Big(\tau, h/\sqrt{\nu_j}\Big) > A_j^\top \lambda(\tau)\Big\} \\
&\quad - \mathbb{I}\Big\{r_j(\tau, \hat{\mu}_j) - h_j\Big(\tau, h/\sqrt{\nu_j}\Big) > A_j^\top \lambda(\tau)\Big\} \\
&\quad = \mathbb{I}\Big\{r_j(\tau, \hat{\mu}_j) - A_j^\top \lambda(\tau) > -h_j\Big(\tau, h/\sqrt{\nu_j}\Big)\Big\} \\
&\quad - \mathbb{I}\Big\{r_j(\tau, \hat{\mu}_j) A_j^\top \lambda(\tau) > +h_j\Big(\tau, h/\sqrt{\nu_j}\Big)\Big\} \\
&\quad = \mathbb{I}\Big\{\Big|r_j(\tau, \hat{\mu}_j) - A_j^\top \lambda(\tau)\Big| \le h_j(\tau)\Big\}. \qquad (4.8)
\end{aligned}$$

In light of Equation (4.7), this suggests the alternate bias correction

$$\frac{1}{n}\sum_{j=1}^{n} \frac{1}{2h\sqrt{\nu_j}}\mathbb{E}\Big[\mathbb{I}\Big(\Big|r_j(\tau, \hat{\mu}_j) - A_j^\top \lambda(\tau)\Big| \le h_j(\tau)\Big)\Big].$$

Estimating the expectation by its sample average (Lemma D.10 in Online Appendix D) and replacing $\lambda(\tau)$ with $\lambda(\tau, \hat{\mu})$ (Lemma D.11 in Online Appendix D) gives the bias-correction term appearing in Equation (4.3).

Note that an advantage of this custom bias-correction term over the explicit finite difference procedure of Theorem 3.5 is that evaluating the bias correction for a fixed τ only involves determining $\lambda(\tau, \hat{\mu})$, which is typically obtained for free as a by-product of computing $x(\tau, \hat{\mu})$. Thus, we need only solve one optimization per τ instead of $2n$.

The detailed proof of Theorem 4.3 in Online Appendix D provides explicit (finite n) bounds on each of the $o_p(1)$ remainder terms to show that they are uniformly small. As an aside, we note that one can also intuitively derive $B^{Bayes}(\tau, h, \hat{\mu})$ by simply plugging in the approximation Equation (4.5) into our original bias-correction $B(\theta, h, \hat{\mu})$ and simplifying along the lines of Equation (4.8).

4.3. Performance in the Large-Sample Regime

Despite the slight differences between $B^{Bayes}(\tau, h, \hat{\mu})$ and our original bias correction $B(\theta, h, \hat{\mu})$, the proofs of Lemma 3.1 and Theorem 3.6 carry through with almost no adjustments, yielding the following.

Corollary 4.5 (Bound to Full-Information Optimum). *Let $\hat{\tau}$ be a solution to Equation (4.3). Then, under Assumption 2.1,*

$$\mathbb{E}\Big[\frac{1}{n}\big|\mu^\top(x^*(\mu) - x(\hat{\tau}, \hat{\mu}))\big|\Big] \le \frac{2 + h^{-1}}{\sqrt{\nu_{\min}}}.$$

Following an argument identical to the one for Corollary 3.7, we can also prove that $x(\hat{\tau}, \hat{\mu})$ achieves full-information optimal performance in the large-sample regime. We omit the details for brevity.

5. Regularization Policies over Polyhedral Feasible Regions

Another approach to improving the finite-sample behavior of SAA is to use a regularizer. We next tailor our framework in Equation (3.3) to our regularization policies introduced in Example 3.3.

$$\begin{aligned}
&\mathscr{X}^{Reg}(\hat{\mu}) \equiv \{x^{\mathscr{R}}(\Gamma, \hat{\mu}) : \Gamma \in [\Gamma_{\min}, \Gamma_{\max}]\}, \\
&\quad \text{where } x^{\mathscr{R}}(\Gamma, \hat{\mu}) \in \arg\max_{x \in \mathscr{X}} \frac{1}{n}\hat{\mu}^\top x - \frac{\Gamma\sqrt{\nu_{\min}}}{2n}\sum_{j=1}^{n}\frac{x_j^2}{\nu_j}.
\end{aligned}$$

Here Γ controls the amount of regularization. Our choice to scale Γ by $\frac{\sqrt{\nu_{\min}}}{2n}$ is without loss of generality but simplifies our large-sample analysis in Section E.5 in Online Appendix E. This regularizer penalizes the SAA objective, discouraging solutions that contain many low-precision components. Choosing low-precision components causes SAA to perform poorly in Example 3. Thus, we intuitively expect regularized solutions with well-chosen Γ to improve on SAA.

We note that $\mathscr{X}^{Reg}(\hat{\mu})$ can equivalently be cast as robust optimal solutions over uncertainty sets given by ellipsoids. See Ben-Tal and Nemirovski (2002) for an overview of robust optimization and Section E.1 in in Online Appendix E for discussion and a formal statement.

5.1. Cross-Validation Approaches to Selecting Γ

The most common data-driven approach to selecting the regularization parameter in the large-sample regime is to use some form of cross-validation (Friedman et al. 2001). Unfortunately, cross-validation procedures

may not be well defined in the small-data, large-scale regime and, even when they are well defined, may not perform comparably to the oracle. We prove this claim for the canonical examples of leave-one-out (LOO) and K-fold cross-validation. Informally, K-fold cross-validation divides the data into K roughly equal portions or *folds* and then iteratively forms the policy class using all the data except the kth fold and evaluates these policies on the left-out fold. It then selects $\Gamma^{\text{K-fold}}$ to maximize the average performance, where averaging is performed over all K possible choices of the left-out fold. In practice, K is typically taken to be 2, 5, or 10. LOO validation is the special case where $K = \mathsf{S}$, and each component has the same amount of data, that is, $\mathsf{S}_j = \mathsf{S}$. Hold-out validation is the special case where $K = 2$, and one maximizes the performance when leaving out the first fold. Notice that K-fold cross-validation is not defined when $K > \mathsf{S}$, and in the small-data, large-scale regime, S may be very small, for example, less than 10, so not all forms of cross-validation may even be defined.

We only analyze K-fold cross-validation in the specific setting of Example 2.2 for the special case that $\mathsf{S}_j = \mathsf{S} \geq K$ for all j, and S is a multiple of K for simplicity. Let

$$\overline{\mu}_j^k \equiv \frac{K}{\mathsf{S}} \sum_{l=(k-1)\mathsf{S}/K+1}^{k\mathsf{S}/K} \hat{\mu}_j^l,$$
$$\overline{\mu}_j^{-k} \equiv \frac{K}{\mathsf{S}(K-1)} \sum_{\substack{1\leq l\leq (k-1)\mathsf{S}/K,\\ k\mathsf{S}/K<l\leq \mathsf{S}}} \hat{\mu}_j^l$$

be sample averages of the jth coordinate in and excluding the kth fold, respectively. Then

$$x^{\mathsf{K}}(\Gamma, \overline{\mu}^{-k}) \in \arg\max_{x\in\mathscr{X}} \frac{1}{n} x^\top \overline{\mu}^{-k} - \frac{\Gamma}{2n}\frac{1}{\sqrt{\mathsf{S}\nu_0}}\sqrt{\frac{K}{K-1}}\sum_{j=1}^n x_j^2$$

is the analogue of Equation (3.2) for this setting that excludes the kth fold and accounts for the adjusted precision. The K-fold cross-validation solution selects

$$\Gamma^{\text{K-fold}} \in \arg\max_{\Gamma\in[\Gamma_{\min},\Gamma_{\max}]} \frac{1}{K}\sum_{k=1}^K \frac{1}{n}\overline{\mu}^{k\top} x^{\mathsf{K}}\left(\Gamma, \hat{\mu}^{-k}\right) \tag{5.1}$$

and implements $x^{\mathscr{R}}(\Gamma^{\text{K-fold}}, \hat{\mu})$. Notice that the implemented policy uses the full data (all folds). The following theorem proves that there exist instances in which 5-fold, 10-fold, and LOO validation do not achieve oracle performance in the small-data, large-scale regime.

Theorem 5.1 (Leave-One-Out and K-Fold Cross-Validation Are Not Best-in-Class). *There exists a sequence of instances $\{(P^n, \hat{\mu}^n, \nu^n) : \hat{\mu}^n \in \mathbb{R}^n, \nu^n \in \mathbb{R}_+^n, n \geq 2\}$ in the small-data, large-scale regime with $\hat{\mu}^n$ as in Example 1 with $\mathsf{S}_j = \mathsf{S} \equiv 10$ for all j such that*

$$\lim_{n\to\infty} \frac{\frac{1}{n}\mu^{n\top} x^{\mathscr{R}}(\Gamma^{\text{K-fold},n}, \hat{\mu})}{\frac{1}{n}\mu^{n\top} x^{\mathscr{R}}(\Gamma^{\text{OR},n}, \hat{\mu})} < 0.03,$$

for any $K \in \{2, 5, 10\}$. Here $\Gamma^{\text{K-Fold},n}$ and $\Gamma^{\text{OR},n}$ are the K-fold and oracle Γ for the nth instance. In other words, none of the 5-fold, 10-fold, hold-out, or LOO validations achieves oracle performance in the small-data, large-scale regime for this instance.

The proof of this theorem is constructive (see the online appendices). Intuitively, K-fold cross-validation does not achieve oracle performance because, when S is small, $x^{\mathsf{K}}(\Gamma, \overline{\mu}^{-k})$ may be very different from $x^{\mathscr{R}}(\Gamma, \hat{\mu})$. For example, if $\mathsf{S} = 2$, leaving out one data point amounts to throwing away 50% of the data. Hence, the right-hand side of Equation (5.1) is a poor approximation to $\mu^\top x^{\mathscr{R}}(\Gamma, \hat{\mu})$. If, in addition, the true oracle curve $\Gamma \mapsto \mu^\top x^{\mathscr{R}}(\Gamma, \hat{\mu})$ is not very flat near its optimum, then this poor approximation generally does not yield a near minimizer. One can see this intuition directly in the proof of Theorem 5.1, in particular, in Figure EC.1 in Online Appendix E.

5.2. A Near-Oracle Policy for the Regularization-Inspired Class

We next use Theorem 3.5 to identify a policy that performs comparably to the oracle. As in Section 4.1, we exploit the structure of the policy class to develop a more computationally efficient bias correction. Lemma E.2 in Online Appendix E shows that the dual problem to Equation (3.2) is

$$\lambda^{\mathscr{R}}(\Gamma, \hat{\mu}) \in \arg\min_{\lambda\geq 0} b^\top\lambda + \frac{1}{n}\sum_{j=1}^n w_j\left(\Gamma, \hat{\mu}_j - A_j^\top\lambda\right),$$

where

$$w_j(\Gamma, t) = \begin{cases} 0, & \text{if} \quad t < 0, \\ \dfrac{\nu_j}{2\Gamma\sqrt{\nu_{\min}}} t^2, & \text{if} \quad 0 \leq t \leq \frac{\Gamma\sqrt{\nu_{\min}}}{\nu_j}, \\ t - \dfrac{\Gamma\sqrt{\nu_{\min}}}{2\nu_j}, & \text{if} \quad \frac{\Gamma\sqrt{\nu_{\min}}}{\nu_j} < t. \end{cases} \tag{5.2}$$

Intuitively, the function $w_j(\Gamma, t)$ is a smoothed approximation of the hinge function $(t - \frac{\Gamma\sqrt{\nu_{\min}}}{\nu_j})^+$. Lemma E.2 in Online Appendix E also shows that $x_j^{\mathscr{R}}(\Gamma, \hat{\mu})$ can be written explicitly in terms of the dual solution as

$$x_j^{\mathscr{R}}(\Gamma, \hat{\mu}) = \frac{\nu_j}{\Gamma\sqrt{\nu_{\min}}}\left(\left[\hat{\mu}_j - A_j^\top\lambda^{\mathscr{R}}(\Gamma, \hat{\mu})\right]^+ - \left[\hat{\mu}_j - A_j^\top\lambda^{\mathscr{R}}(\Gamma, \hat{\mu}) - \frac{\Gamma\sqrt{\nu_{\min}}}{\nu_j}\right]^+\right).$$

Substituting this expression into our standard bias correction and taking the limit as $h \to 0$ suggests the alternate, more computationally efficient correction

$$B_n^{Reg}(\Gamma, \hat{\mu}) \equiv \frac{1}{\Gamma n \sqrt{\nu_{\min}}} \sum_{j=1}^{n} \mathbb{I}\left(0 \le \hat{\mu}_j - \boldsymbol{A}_j^\top \boldsymbol{\lambda}^{\mathcal{R}}(\Gamma, \hat{\mu}) \le \frac{\Gamma\sqrt{\nu_{\min}}}{\nu_j}\right).$$

Finally, for any $0 < \Gamma_{\min} < \Gamma_{\max} < \infty$, define

$$\hat{\Gamma} \in \arg\max_{\Gamma_{\min} \le \Gamma \le \Gamma_{\max}} \frac{1}{n} \hat{\mu}^\top \boldsymbol{x}^{\mathcal{R}}(\Gamma, \hat{\mu}) - B_n^{Reg}(\Gamma, \hat{\mu}). \tag{5.3}$$

Under assumptions similar to Theorem 4.3, we can strengthen Theorem 3.5 for $\boldsymbol{x}^{\mathcal{R}}(\hat{\Gamma}, \hat{\mu})$ by bounding the maximal stochastic deviations.

Theorem 5.2 (Finite-Dimension Bound for the Regularization Policy Class). *Consider P^n under Assumption 4.2 with $\mathcal{X}$ as in Equation (4.1) with $m \ge 1$, and $\mathcal{X}$ is s_0-strictly feasible. Let C_μ, C_A be such that $|\mu_j| \le C_\mu$ and $\|\boldsymbol{A}_j\|_\infty \le C_A$ for all $j = 1, \ldots, n$, and $\beta > 0$ be such that the minimum eigenvalue of $\frac{1}{n}\sum_{j=1}^n \boldsymbol{A}_j \boldsymbol{A}_j^\top \in \mathbb{R}^{m\times m}$ is at least β. Assume that $0 < \Gamma_{\min} < \Gamma_{\max} < \infty$. Then, for each $0 < \delta < 1$, there exist positive constants C_1, C_2, C_3, C_4 not depending on $\{n, m, \delta\}$ such that*

$$0 \le \frac{1}{n}\boldsymbol{\mu}^\top \left(\boldsymbol{x}^{\mathcal{R}}\left(\Gamma^{\mathsf{OR}}, \hat{\mu}\right) - \boldsymbol{x}^{\mathcal{R}}\left(\hat{\Gamma}, \hat{\mu}\right)\right) \le \underbrace{C_1\delta}_{\substack{\text{Approximating}\\ \text{Dual Solution}}} + \underbrace{C_1 \mathsf{TV}\log(e/\mathsf{TV})}_{\substack{\text{Degree}\\ \text{NonNormality}}} + \underbrace{2R_n}_{\substack{\text{Stochastic}\\ \text{Errors}}},$$

where, for any $\epsilon > \delta^2/\sqrt{n}$,

$$\mathbb{P}\{R_n > 4\epsilon\} \le C_2\left(\exp\left(\frac{-C_3\delta\sqrt{n}}{\sqrt{m}\log(m+1)}\right) \times \log\left(1 + \frac{\sqrt{m}\log(m+1)}{C_3\delta\sqrt{n}}\right) + \exp\left(\frac{-C_4\epsilon\sqrt{n}}{\log(m+1)}\right)\right), \tag{5.4}$$

and $\mathsf{TV} = \frac{1}{2n}\sum_{j=1}^n \|\phi_j - \phi\|_1$ is the average total variation distance between $\phi_j(\cdot)$ and the standard normal density $\phi(\cdot)$.

Like Theorem 4.3, Theorem 5.2 holds for a fixed instance $(P^n, \hat{\mu}, \boldsymbol{\nu})$, that is, finite n. The structure of the bound, however, provides insight into the performance in the small-data, large scale regime and shows that the performance of our policy converges to that of the oracle when the dimension increases and $\hat{\mu}_j$ is Gaussian. This result is stated in the following corollary.

Corollary 5.3 (Almost-Sure Convergence to Oracle). *Consider a sequence of instances of $\{(P^n, \hat{\mu}^n, \boldsymbol{\nu}^n) : n \ge 2\}$ in the small-data, large-scale regime with each instance satisfying Assumption 4.2 with $\mathcal{X}_n$ given by Equation (4.1) and each $\mathcal{X}_n$ is s_0-strictly feasible (see Definition 4) for a common s_0. Suppose further that the parameters do not grow with n, that is, m is constant, $\|\boldsymbol{A}_j\| \le C_A$, $|\mu_j^n| \le C_\mu$, and $\nu_{\min} \le \nu_j^n \le \nu_{\max}$ for all $1 \le j \le n$ and that the smallest eigenvalue of $\frac{1}{n}\sum_{j=1}^n \boldsymbol{A}_j^n \boldsymbol{A}_j^{n\top} \in \mathbb{R}^{m\times m}$ is at least β for all n. For each P_n, let $\hat{\Gamma}^n$ be given by Equation (5.3). If each $\hat{\mu}_j$ is Gaussian, then the policy $\boldsymbol{x}^{\mathcal{R}}(\hat{\Gamma}^n, \hat{\mu})$ performs as well as the oracle policy for $\mathcal{X}^{Reg,n}(\hat{\mu})$ a.s. as $n \to \infty$.*

We stress that it is unclear a priori whether $\boldsymbol{x}(\hat{\tau}, \hat{\mu})$ or $\boldsymbol{x}^{\mathcal{R}}(\hat{\Gamma}, \hat{\mu})$ yields better performance in problem P^n because it is not clear which benchmark, $\boldsymbol{x}(\tau^{\mathsf{OR}}, \hat{\mu})$ or $\boldsymbol{x}^{\mathcal{R}}(\Gamma^{\mathsf{OR}}, \hat{\mu})$, is superior. In general, we find that the difference is application specific.

As in Theorem 4.3, the proof of Theorem 5.2 also follows that of Theorem 3.5 and uses an average dual problem to break the dependence between terms (Lemma E.6 in Online Appendix E). Practically, the restriction $0 < \Gamma_{\min} < \Gamma_{\max} < \infty$ is mild; we expect in practice to optimize $\hat{\Gamma}$ by searching over a finite grid. It is an open question whether Theorem 5.2 can be strengthened to allow $\Gamma_{\min} = 0$ or $\Gamma_{\max} = \infty$.

Although we focused on the small-data, large-scale regime, our policy also achieves full-information optimality in the large-sample regime. (Note that Theorem 3.6 is not applicable when $\Gamma_{\min} > 0$ because $\boldsymbol{x}^{SAA}(\hat{\mu})$ may not be in $\mathcal{X}^{Reg}(\hat{\mu})$). See Section E.5 in Online Appendix E for the theorem and discussion.

6. Numerical Experiments

We next study the empirical performance of our methods in the context of a specific application: the online advertising portfolio optimization problem (OAPOP). Our goals are twofold: (1) quantify the potential benefits of our Bayes- and regularization-inspired best-in-class policies over traditional variants for this application and (2) assess the robustness of our results to increasingly large departures from normality.

We focus on the OAPOP because we view it as typical of the small-data, large-scale optimization regime. Loosely speaking, the OAPOP involves an advertiser seeking to allocate a finite budget among various *targeting items* to maximize the advertiser's return. Targeting items may represent keywords, impressions, cookies, and websites and may span different advertising channels and platforms. In practice, an advertiser must also estimate the expected cost and revenue for each targeting item before electing an allocation. Pani et al. (2017) provide a thorough overview of the OAPOP, including its pivotal role in the online advertising industry and the recent surge of decision-support software products for the problem, such as Adobe Marketing Cloud and Google's DoubleClick. Most important, the authors observe that a typical instance of the OAPOP may involve tens of thousands of targeting items and that, because the underlying

problem parameters shift rapidly, estimates are necessarily very noisy—the two defining features of the small-data, large-scale regime.

Pani et al. (2017) argue that despite the many complexities of the online advertising industry, the OAPOP can be modeled effectively as an off-line, fractional, multichoice knapsack problem. The choices correspond to different bid levels for each targeting item, and the weights and rewards correspond to the expected costs and revenues. The authors propose a customized algorithm for massively large instances in which the expected returns and expected cost of each item are known. We adopt a similar perspective, complementing their work by focusing instead on instances in which the returns are not known but rather imprecisely estimated. As in Rusmevichientong and Williamson (2006), we consider only a single, representative bid level. The resulting problem is an off-line, fractional knapsack problem with uncertain rewards. It can be written in the form of problem P^{n} with $\mathscr{X} = \{\boldsymbol{x} \in [0,1]^n : \frac{1}{n}\boldsymbol{c}^\top \boldsymbol{x} \le 1\}$ for some fixed-cost vector $\boldsymbol{c} \in \mathbb{R}^n$.

We simulate a variety of instances of the OAPOP and apply our data-driven procedures. Our precise simulation procedure for $\boldsymbol{\mu}$, $\hat{\boldsymbol{\mu}}$, and $\boldsymbol{\nu}$ is in Section F.2 in Online Appendix F and closely follows the procedure of Pani et al. (2017). Those authors calibrated this procedure to match a real-world data set drawn from Google's Keyword Planner tool for "medium-high volume keywords from a wide variety of industry categories including retail (apparel, footwear, etc.), insurance, and financial services" (Pani et al. 2017, p. 23). Overall, although simulated, we believe our instances to be realistic in structure.

Throughout our experiments, SAA is the sample average approximation policy; EB OR is the empirical Bayes' oracle policy $\boldsymbol{x}(\tau^{\mathrm{OR}}, \hat{\boldsymbol{\mu}})$; EB OPT is our proposed policy $\boldsymbol{x}(\hat{\tau}, \hat{\boldsymbol{\mu}})$; EB MLE and EB MM are the estimate-then-optimize policies based on empirical Bayes' maximum likelihood estimate (MLE) and moment-matching (MM) estimate, respectively; and SURE is the estimate-then-optimize policy based on the SURE estimate (cf. Section 4). Similarly, Reg OR is the oracle policy $\boldsymbol{x}^{\mathscr{R}}(\Gamma^{\mathrm{OR}}, \hat{\boldsymbol{\mu}})$, and Reg OPT is our proposed policy $\boldsymbol{x}^{\mathscr{R}}(\hat{\Gamma}, \hat{\boldsymbol{\mu}})$. RO 1% and RO 5% are robust optimization (RO) policies using elliptical uncertainty sets for two different choices of r (cf. Section E.1 in Online Appendix E). Specifically, we use the *safe-approximation guideline* for robustness (Ben-Tal and Nemirovski 2002, chapter 2), and for $\epsilon = 0.01, 0.05$, we let $r = \sqrt{2\log(1/\epsilon)}$. For both the Bayes- and regularization-inspired classes, we also compare with holdout (HO), fivefold (K5), and LOO cross-validation. We note that cross-validation strategies are seemingly rare in the empirical Bayes literature but can be defined analogously as in the regularization case. We also remind the reader that K-fold cross-validation is not well defined when $S < K$.

The code for all experiments written in the Julia programming language (Bezanson et al. 2017) is available at https://github.com/vgupta1/EmpBayes. When computing $\hat{\tau}^n$, we take $h_n = n^{-1/6}$ and search exhaustively over $\tau \in [0,5]$ using a grid of size 0.01. When computing τ^{OR}, we use a parametric linear programming algorithm to find the exact optimum over $\mathbb{R}_+$. Similarly, when computing both $\hat{\Gamma}$ and Γ^{OR}, we search exhaustively over $\Gamma \in [1,100]$ using a grid of size 0.5. No special effort was devoted to tuning these parameters.

Before proceeding, we summarize our main insights as follows:

1. For large n, our small-data, large-scale methods (EB OPT, Reg OPT) offer significant benefits over SAA. They also offer a smaller benefit over estimate-then-optimize methods that do not leverage the optimization structure.

2. For this particular application, cross-validation methods perform well, often comparable to the oracle, in contrast to their theoretical analysis. This distinction arises because the OAPOP is very flat around its optimum.

3. For small to moderate n, our methods exhibit somewhat more variability than estimate-then-optimize methods and, as a consequence, can have poorer performance. Their variability, however, is comparable with that of cross-validation procedures.

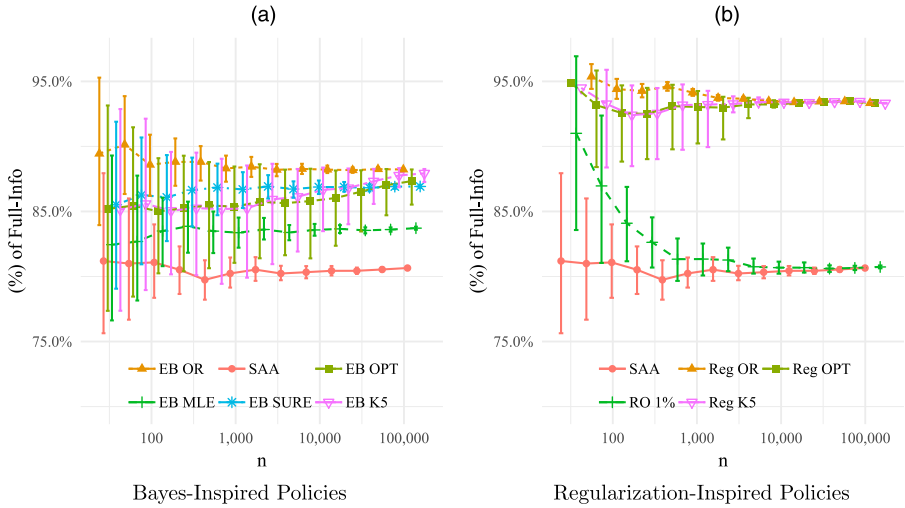
4. Overall, the performance of our methods seems robust to mild departures from normality.

6.1. Finite-Dimensional Behavior (Finite n)

We first study the behavior of our approach as n grows when $\hat{\mu}_j$ is Gaussian, that is, $\mathsf{TV} = 0$. Figure 4 shows the performance of various data-driven methods across 200 simulations for increasing n. We have split the methods into two plots—Bayes- and regularization-inspired policies—and included a subset of policies for readability. A larger plot with all policies is in Section F.3 in Online Appendix F.

With respect to the Bayes-inspired policies, several features are immediately clear. First, as expected, our proposed policy EB OPT offers significantly better performance than estimate-then-optimize policies and SAA, especially as n grows large. By contrast, cross-validation procedures, which do leverage the optimization structure, have very similar performance in this example. A drawback of both EB OPT and the cross-validation procedures is that they are more variable than competitors. This feature is more pronounced in Figure 5, in which we plot the standard deviation of the performance (relative to the

Figure 4. (Color online) Relative Performance by Policy for OAPOP



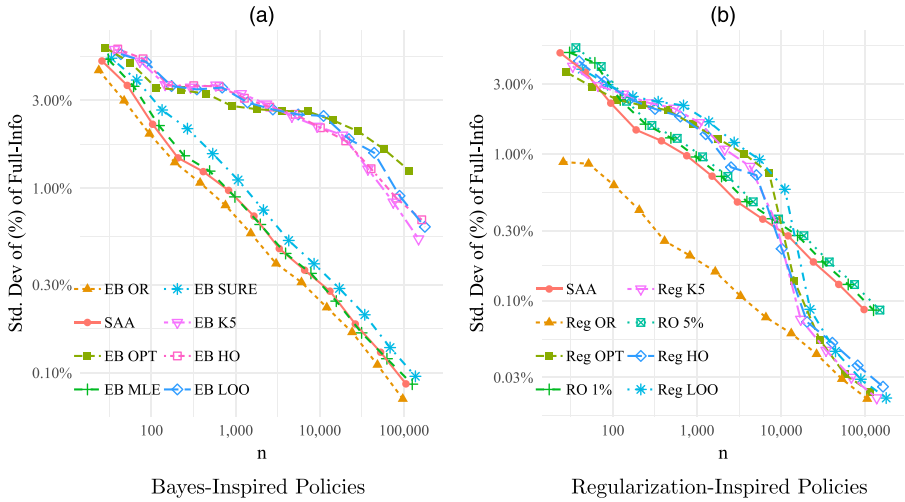
Notes. The left panel plots the performance of $x(\tau, \hat{\mu})$ for various n and data-driven procedures for choosing τ from Section 4. The right panel plots the performance of $x^{\mathcal{R}}(\Gamma, \hat{\mu})$ along the same sample paths for data-driven procedures for choosing Γ from Section 5. The error bars represent 10% and 90% quantiles over 200 simulations.

full-information optimum) of each method along the 200 simulations. Nonetheless, from Figure 4, we would argue that the benefits in average performance outweigh the extra variability for $n > 30,000$, a fairly reasonable number for this application.

Regularization-inspired policies exhibit similar performance. Our proposed policy Reg OPT converges quickly to oracle performance. Cross-validation approaches are highly competitive. By contrast, the robust optimization policies with radii specified according to the safe-approximation guideline have significantly worse performance, converging to SAA as $n \to \infty$.

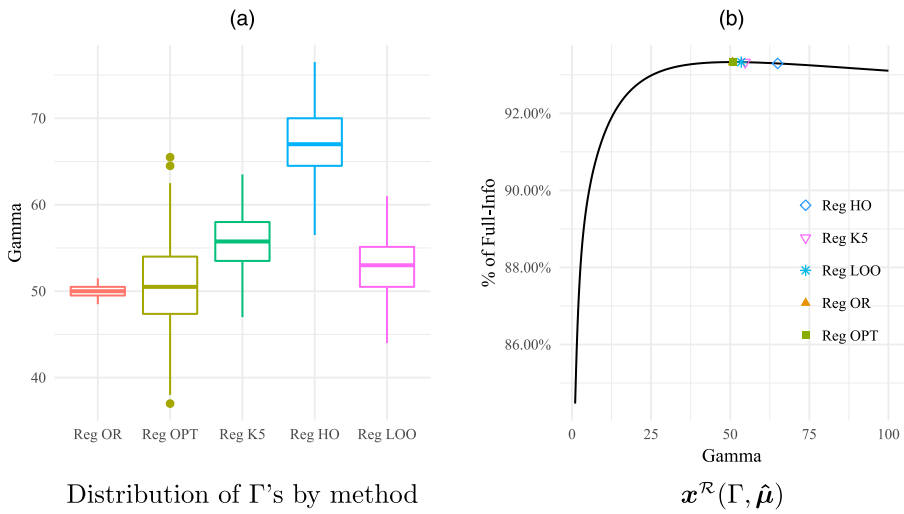
In the case of regularization-inspired policies, the strong performance of cross-validation sharply contrasts with Theorem 5.1. We believe that this performance is somewhat application specific. Recall that cross-validation policies perform poorly when (1) the cross-validation estimates for each fold differ greatly from the original policy class (i.e., $x^k(\Gamma, \overline{\mu}^{-k})$ differs from $x^{\mathcal{R}}(\Gamma, \hat{\mu})$ in Equation (5.1)) and (2) the oracle curve $\Gamma \mapsto x^{\mathcal{R}}(\Gamma, \hat{\mu})$ is not too flat at its optimum. For our OAPOP instances, the first condition holds. This is somewhat difficult to see by examining the curves directly (left panel of Figure EC.5 in Online Appendix F), but it is more evident by noting that the

Figure 5. (Color online) Standard Deviation of Performance for OAPOP



Notes. The left panel plots the standard deviation of $x(\tau, \hat{\mu})$ for various n and data-driven procedures for choosing τ from Section 4. The right panel plots the standard deviation of $x^{\mathcal{R}}(\Gamma, \hat{\mu})$ along the same sample paths for data-driven procedures for choosing Γ from Section 5. Note the log-log scales.

Figure 6. (Color online) Explaining Performance of Cross-Validation for OAPOP



Notes. The left panel plots the distribution of the optimizing Γ for various methods across the 200 simulations. The right panel plots the oracle curve $\Gamma \to \mu^{\top} x^{\mathcal{R}}(\Gamma, \hat{\mu})$ for a single realization when $n = 2^{17}$, with optimizing Γ of other methods indicated.

optimizing Γ values for these curves converge to different values as $n \to \infty$ (right panel of Figure EC.5 in Online Appendix F). Indeed, Figure 6 shows boxplots of the distribution of the various estimated Γ values when $n = 2^{17}$, highlighting that the cross-validation procedures overregularize in this example, and Reg Opt converges to Γ^{OR}.

The key observation, however, is that the second condition does not hold for these instances: the oracle curve is very flat (see the right panel of Figure 6). Hence, even though cross-validation procedures select the wrong Γ, this error manifests as a neglible amount of suboptimality. (Contrast this curve with the oracle curve in Theorem 5.1 seen in Figure EC.1 in Online Appendix E.) In our opinion, this feature explains the strong performance of *K*-fold cross-validation in this application, and Theorem 5.1 shows that this feature does not always hold. By contrast, our Reg Opt policy is guaranteed to achieve oracle performance in the small-data, large-scale regime under the conditions in Theorem 5.2.

A very similar analysis can be performed for the cross-validation policies in the Bayes-inspired policy class. See Figures EC.6 and EC.7 in Online Appendix F for the details. There, too, we notice that the oracle curve is quite flat at its optimum (in contrast to, e.g., Example 4.1), partially explaining the strong performance of cross-validation for these instances.

As an aside, we note that the optimal policies EB OPT and Reg OPT typically have very different structures. For example, on a typical path with $n = 2^{17}$, more than 50% of the positive values of $x^{\mathcal{R}}(\hat{\Gamma}, \hat{\mu})$ (Reg OPT) are fractional. By contrast, $x(\hat{\tau}, \hat{\mu})$ (EB OPT) has at most one fractional value by construction regardless of the size of n. Depending on the intended application, this difference in structure may be pertinent.

6.2. Other Experiments

Sections F.4 and F.5 in Online Appendix F present additional experiments assessing the relative performance of our methods when S is large but finite and when $\hat{\mu}$ is non-Gaussian, respectively. Generally, we find that EB Opt and Reg Opt are competitive with cross-validation methods and generally outperform SAA and estimate-then-optimize methods when S is large even when estimators are non-Gaussian. In particular, our method is robust to some non-normality when noise is still sub-Gaussian and admits a density (see the online appendices for details).

7. Conclusion and Future Directions

Motivated by emerging optimization applications in which the amount of relevant data per parameter is small, we propose a novel method for dealing with linear optimization in the small-data, large-scale regime. In contrast to the large-sample regime, the full-information optimum is not achievable in this context, so we must focus on finding a best-in-class policy. As a benchmark, we consider an oracle policy that knows the underlying parameters in advance but is constrained to use a policy from a specific class.

By correcting for the bias in the estimated objective value, we developed a novel framework for designing a policy whose performance converges to this oracle benchmark and applied the framework to two important classes of policies: Bayes- and

regularization-inspired policies. Numerical results show that our policies are robust and perform well across a broad range of problem instances, surpassing traditional methods in the literature.

Our framework can be applied to an arbitrary class of policies. However, to ensure convergence, we need to establish uniform convergence properties of the bias-correction terms. It would be interesting to explore other classes of policies and optimization problems for which such uniform convergence can be established. This paper focused on linear optimization, but the spirit of our approach can be extended to nonlinear and discrete optimizations as well. This is a challenging but potentially exciting research area. It is our hope that this research galvanizes the community to consider optimization problems within this important small-data, large-scale regime.

Acknowledgments

The authors thank Area Editor Professor Yinyu Ye, the associate editor, and the three referees for their detailed and thoughtful comments, which substantially improved the exposition and technical results.

Endnotes

[1] The majority of the literature treats an expected value of objective as in Equation (1.1), often because the selected solution is held fixed for a long time. Alternatively, one could study the out-of-sample performance as random variable, that is, $c(\hat{x}, \bar{\xi})$, where $\hat{x}$ depends on the data and $\bar{\xi}$ is an i.i.d. copy of ξ. For problem P_n with n large, however, this out-of-sample performance is an average of n terms and hence converges to its expectation whenever these terms are sufficiently independent by the strong law of large numbers, further motivating our focus on expected values.

[2] As an aside, we mention that by the central limit theorem, we expect this estimator to be approximately Gaussian if the ξ_j^k are not too heavy tailed and S_j is not too small.

[3] We focus on $n \geq 2$ to avoid some trivial cases.

[4] As an aside, we mention that Frees (1991) proves that under mild assumptions, the $\hat{\mu}_j$ in this model are approximately normally distributed.

[5] Recall that a mean-zero random variable ξ is sub-Gaussian with variance proxy σ^2 if $\mathbb{E}[\exp(t\xi)] \leq \exp(t^2\sigma^2/2)$ for all $t \in \mathbb{R}$ (Wainwright 2019). Estimators are frequently modeled as sub-Gaussian.

[6] We adopt the usual convention that $0 \cdot \log 0 = 0$ by continuity.

[7] For example, standard central limit theorem results still hold when the summands are mildly dependent. Intuitively, similar phenomena should hold here.

[8] Of course, because we treat $\boldsymbol{\mu}$ as an unknown constant and not a realization from a prior in our analysis, there is technically no notion of a Bayes-optimal policy, hence necessitating the qualifier *inspired*.

[9] For clarity, estimate-then-optimize policies are policies computed by first estimating $\boldsymbol{\mu}$ using some statistical criteria (e.g., maximum likelihood estimation) and then plugging in that estimator for $\boldsymbol{\mu}$ in P^n and solving.

[10] If the equation has no solution, take $\tau^{MLE} = 0$.

[11] If the equation has no solution, take $\tau^{SURE} = 0$.

[12] For example, if $\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}$ encoded an inequality constraint, for example, $\boldsymbol{a}_0^\top \boldsymbol{x} \leq \boldsymbol{b}_0$ and $-\boldsymbol{a}_0^\top \boldsymbol{x} \leq -\boldsymbol{b}_0$, we might relax these constraints to $\boldsymbol{a}_0^\top \boldsymbol{x} \leq \boldsymbol{b}_0 + \delta$ and $-\boldsymbol{a}_0^\top \boldsymbol{x} \leq -\boldsymbol{b}_0 + \delta$ to achieve strict feasibility. The extent to which such a relaxation is an acceptable approximation is application dependent.

References

Ban G-Y, Rudin C (2014) The big data newsvendor: Practical insights from machine learning. *Oper. Res.* 67(1):90–108.

Ban G-Y, Gallien J, Mersereau A (2018) Dynamic procurement of new products with covariate information: The residual tree method. *Manufacturing Service Oper. Management* 21(4):798–815.

Belloni A, Chernozhukov V (2011) ℓ_1-penalized quantile regression in high-dimensional sparse models. *Ann. Statist.* 39(1):82–130.

Ben-Tal A, Nemirovski A (2000) Robust solutions of linear programming problems contaminated with uncertain data. *Math. Programming* 88(3):411–424.

Ben-Tal A, Nemirovski A (2002) Robust optimization–methodology and applications. *Math. Programming* 92(3):453–480.

Berger JO (2013) *Statistical Decision Theory and Bayesian Analysis* (Springer-Verlag, New York).

Bertsimas D, Kallus N (2019) From predictive to prescriptive analytics. *Management Sci.*, ePub ahead of print August 23, https://pubsonline.informs.org/doi/10.1287/mnsc.2018.3253.

Bertsimas D, Sim M (2004) The price of robustness. *Oper. Res.* 52(1):35–53.

Bertsimas D, Tsitsiklis JN (1997) *Introduction to Linear Optimization* (Athena Scientific, Belmont, MA).

Bertsimas D, Gupta V, Kallus N (2018a) Data-driven robust optimization. *Math. Programming* 167(2):235–292.

Bertsimas D, Gupta V, Kallus N (2018b) Robust sample average approximation. *Math. Programming* 171(1–2):217–282.

Bezanson J, Edelman A, Karpinski S, Shah VB (2017) Julia: A fresh approach to numerical computing. *SIAM Rev.* 59(1):65–98.

Bickel PJ, Ritov Y, Tsybakov AB (2009) Simultaneous analysis of Lasso and Dantzig selector. *Ann. Statist.* 37(4):1705–1732.

Bousquet O, Elisseeff A (2002) Stability and generalization. *J. Machine Learn. Res.* 2(3):499–526.

Bühlmann P, Geer SA (2011) *Statistics for High-Dimensional Data* (Springer-Verlag, Berlin).

Candès EJ, Recht B (2009) Exact matrix completion via convex optimization. *Foundations Comput. Math.* 9(6):717–772.

Candès EJ, Sing-Long CA, Trzasko JD (2013) Unbiased risk estimates for singular value thresholding and spectral estimators. *IEEE Trans. Signal Processing* 61(19):4643–4657.

Chen X, Sim M, Sun P (2007) A robust optimization perspective on stochastic programming. *Oper. Res.* 55(6):1058–1071.

Chen X, Owen Z, Pixton C, Simchi-Levi D (2015) A statistical learning approach to personalization in revenue management. Working paper, Massachussetts Institute of Technology, Cambridge.

Delage E, Ye Y (2010) Distributionally robust optimization under moment uncertainty with application to data-driven problems. *Oper. Res.* 58(3):595–612.

Donoho DL, Johnstone IM (1995) Adapting to unknown smoothness via wavelet shrinkage. *J. Amer. Statist. Assoc.* 90(432):1200–1224.

Efron B (2012) *Large-scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction* (Cambridge University Press, Cambridge, UK).

Efron B, Morris C (1973) Stein's estimation rule and its competitors—An empirical Bayes approach. *J. Amer. Statist. Assoc.* 68(341):117–130.

Efron B, Morris C (1975) Data analysis using Stein's estimator and its generalizations. *J. Amer. Statist. Assoc.* 70(350):311–319.

Elmachtoub AN, Grigas P (2017) Smart "predict, then optimize." Working paper, Columbia University, New York.

Esfahani PM, Kuhn D (2018) Data-driven distributionally robust optimization using the Wasserstein metric: Performance guarantees and tractable reformulations. *Math. Programming* 171(1–2):115–166.

Ferreira KJ, Lee BHA, Simchi-Levi D (2015) Analytics for an online retailer: Demand forecasting and price optimization. *Manufacturing Service Oper. Management* 18(1):69–88.

Frees EW (1991) Linear regression and U-statistics. *Sankhya Ser. A* 53(1):84–96.

Friedman J, Hastie T, Tibshirani R (2001) *The Elements of Statistical Learning* (Springer-Verlag, New York).

Gupta V (2019) Near-optimal Bayesian ambiguity sets for distributionally robust optimization. *Management Sci.* 65(9): 3949–4450.

Johnstone IM (2015) Gaussian estimation: Sequence and wavelet models. Working paper, Stanford University, Stanford, CA.

Kleywegt AJ, Shapiro A, Homem-de Mello T (2002) The sample average approximation method for stochastic discrete optimization. *SIAM J. Optim.* 12(2):479–502.

Lan G (2012) An optimal method for stochastic composite optimization. *Math. Programming* 133(1):365–397.

Lim AEB, Shanthikumar JG, Vahn GY (2011) Conditional value-at-risk in portfolio optimization: Coherent but fragile. *Oper. Res. Lett.* 39(3):163–171.

Liyanage LH, Shanthikumar JG (2005) A practical inventory control policy using operational statistics. *Oper. Res. Lett.* 33(4):341–348.

Morris CN (1983) Parametric empirical Bayes inference: Theory and applications. *J. Amer. Statist. Assoc.* 78(381):47–55.

Mukherjee G, Brown LD, Rusmevichientong P (2015) Empirical Bayes prediction for the multivariate newsvendor loss function. Working paper, University of Southern California, Los Angeles.

Negahban S, Yu B, Wainwright MJ, Ravikumar PK (2012) A unified framework for high-dimensional analysis of *m*-estimators with decomposable regularizers. *Statist. Sci.* 27(4):538–557.

Nemirovski A, Juditsky A, Lan G, Shapiro A (2009) Robust stochastic approximation approach to stochastic programming. *SIAM J. Optim.* 19(4):1574–1609.

Nesterov Y (2005) Smooth minimization of non-smooth functions. *Math. Programming* 103(1):127–152.

Nesterov Y (2009) Primal-dual subgradient methods for convex problems. *Math. Programming* 120(1):221–259.

Pani A, Raghavan S, Sahin M (2017) Large-scale advertising portfolio optimization in online marketing. Working paper, Adobe Systems Inc., San Jose, CA.

Pollard D (1990) *Empirical Processes: Theory and Applications,* NSFCBMS Regional Conference Series in Probability and Statistics, vol. 2 (Institute of Mathematical Statistics, Hayward, CA).

Ross N (2011) Fundamentals of Stein's method. *Probab. Surveys* 8: 210–293.

Rusmevichientong P, Williamson DP (2006) An adaptive algorithm for selecting profitable keywords for search-based advertising services. Feigenbaum J, Chuang J, Pennock DM, eds. *Proc. 7th ACM Conf. Electronic Commerce* (ACM, New York), 260–269.

Ruszczynski AP, Shapiro A (2003) *Stochastic Programming*, vol. 10 (Elsevier, Amsterdam).

Shapiro A, Dentcheva D, Ruszczyński A (2009) *Lectures on Stochastic Programming: Modeling and Theory* (Society for Industrial and Applied Mathematics, Philadelphia).

Stein CM (1981) Estimation of the mean of a multivariate normal distribution. *Ann. Statist.* 9(6):1135–1151.

Tibshirani RJ, Taylor J (2012) Degrees of freedom in lasso problems. *Ann. Statist.* 40(2):1198–1232.

Van der Vaart AW (2000) *Asymptotic Statistics*, Cambridge Series in Statistical and Probabilistic Mathematics, vol. 3 (Cambridge University Press, Cambridge, UK).

Vapnik VN (1999) An overview of statistical learning theory. *IEEE Trans. Neural Networks* 10(5):988–999.

Wainwright MJ (2009) Information-theoretic limits on sparsity recovery in the high-dimensional and noisy setting. *IEEE Trans. Inform. Theory* 55(12):5728–5741.

Wainwright MJ (2019) *High-Dimensional Statistics: A Non-Asymptotic Viewpoint,* Cambridge Series in Statistical and Probabilistic Mathematics, vol. 48 (Cambridge University Press, Cambridge, UK).

Xie X, Kou SC, Brown LD (2012) Sure estimates for a heteroscedastic hierarchical model. *J. Amer. Statist. Assoc.* 107(500):1465–1479.

Zhang C-H (2003) Compound decision theory and empirical Bayes methods. *Ann. Statist.* 31(2):379–390.

Copyright 2021, by INFORMS, all rights reserved. Copyright of Management Science is the property of INFORMS: Institute for Operations Research and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.